



天津职业技术师范大学

Tianjin University of Technology and Education

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基于云平台的 STM 32 微控制器智能家居系统设计

**The Design of STM 32 Micro-Controller Intelligent
Home System Based on Cloud Platform**

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中文摘要

随着科学技术的发展,电子信息和物联网技术应用范围越来越广。随着居民对安全、舒适、便捷等高品质生活的真实需求,市场上不断涌现出众多智能家居相关设备和产品。对整个行业的市场现状和发展前景进行研究分析,可以看出国内市场对于智能家居标准尚未统一,其产品和设备也存在价格高安装使用价格昂贵,维修不便,维护成本高等劣势。这也使得消费者选择使用智能家居存在一定的阻碍。

该系统以 STM32单片机为主控制器,通过多个功能模块的数据采集实现环境监测,微控制器从传感器获取信息并通过远程通信将其传输到云平台。通过温湿度传感器 DHT11采集的数据来监测室内的温湿度,在 OLED 屏幕显示所测数据;通过 MQ2气敏传感器对室内空气中的天然气含量检测;通过火焰/火灾传感器 LM393 模块感应火焰的存在并做出反应。本设计使用的无线传输模块为 ESP8266模块。ESP8266EX 提供完整且自成体系的 Wi-Fi 网络解决方案;它可用于托管应用程序或从另一个应用程序处理器卸载 Wi-Fi 网络功能。继电器用于保护电气系统,并减少由于系统内发生的过电流/电压或其他异常情况而对系统内连接的设备造成的伤害。通过云平台工具连接网络后,经主控判断上传数据,将报警信号和各项数据上传至云平台后,用户可以通过网页端或者手机端查看信息,并发出指令传输到主控完成各项功能的调整。

该系统经过实践检验,性能稳定可靠。在给用户提供方便的同时,也实现了系统的智能化、自动化管理。

关键词: ARM Cortex-M3 STM32微控制器模块,温湿度传感器模块,烟雾传感器模块,无线传输,火焰传感器,蜂鸣器。

Abstract

With the development of science and technology, the application range of electronic information and Internet of Things technology is becoming increasingly extensive. With the real needs of residents for high-quality life such as safety, comfort, and convenience, many smart home-related equipment and products have emerged in the market. The research and analysis of the market status quo and development prospects of the entire industry can be seen that the domestic market has not yet unified the standards for smart homes, and its products and equipment also have the disadvantages of high prices, expensive installation and use, inconvenient maintenance, and high maintenance costs. This also makes consumers choose to use smart homes there are certain obstacles.

The system is based on the STM32 microcontroller, through the data acquisition of multiple functional modules to achieve environmental monitoring, the microcontroller obtains information from the sensor and transmits it to the cloud platform through remote communication. The temperature and humidity in the room is monitored by the data collected by the temperature and humidity sensor DHT11, and the measured data is displayed on the OLED screen; the natural gas content in the indoor air is detected by the MQ2 gas sensor; the presence of the flame is sensed and reacted by the flame/fire sensor LM393 module. The wireless transmission module used in this design is the ESP8266 module. ESP8266EX provides a complete and self-contained Wi-Fi networking solution that can be used to host applications or offload Wi-Fi network functionality from another application processor. Relays are used to protect the electrical system and reduce damage to connected devices within the system due to overcurrent/voltage or other anomalies that occur within the system. After connecting to the network through the cloud platform tool, the upload data is judged by the master control, and the alarm signal and various data are uploaded to the cloud platform, the user can view the information through the web page or mobile phone, and issue instructions to the master to complete the adjustment of various functions.

The system has been evaluated in practice and has stable and reliable performance. While bringing convenience to users, it also realizes the intelligent and automated management of the system.

Keywords: *ARM Cortex-M3 STM32 Micro-controller module, Temperature and humidity sensor module, Smoke sensor module, Wireless transmission, Flame sensor, Buzzer*

目录

Contents

Contents	vi
LIST OF FIGURES	x
Chapter 1 introduction	1
1.1 Overview	1
1.2 History of home security systems	2
1.3 Research significance of the project.	5
Chapter 2 the Design of Hardware.....	6
2.1 Introduction of the system.	6
2.2 STM32F103 Microcontroller.	7
2.3 Temperature and humidity sensor module.	7
2.4 IR Sensor module.	8
2.5 Flame sensor module.	8
2.6 Gas sensor module.....	8
2.7 Buzzer module.	8
2.8 Wireless transmission module.....	8
2.9 Power supply module.	9
Chapter 3 the Design of Hardware Module.....	10
3.1 Introduction to microcontroller	10
3.2 ARM Cortex-M3 STM32F103 micro-controller	12
3.2.1. All features	12
3.2.2 Clock circuit	14
3.2.3 Reset circuit.	16
3.2.4 MCU block diagram.....	17
3.2.5 Pin description of STM32F103C8T6	18
3.3 Temperature and humidity sensor module.	20
3.3.1 The importance of temperature and humidity sensors.....	20
3.3.2 Introduction of DHT 11 humidity & temperature sensor.	21
3.3.3 Technical Specifications:	22
3.3.4 Detailed Specifications:.....	22
3.3.5 Typical Application.	23
3.3.6 Power and Pin.	23
3.3.7 Communication Process: Serial Interface (Single-Wire Two-Way)	23
3.3.8 Overall Communication Process.	23

3.3.9 MCU Sends out Start Signal to DHT	24
3.3.10 DHT Responses to MCU	24
3.3.11 Electrical Characteristics	25
3.3.12 Attentions of application	25
3.4 IR Infrared Obstacle Avoidance Sensor Module.	26
3.4.1 Working principle.....	26
3.4.2 Features of IR Sensor Module.....	27
3.4.3 Connections.	27
3.4.4 Pin Description.	28
3.4.5 Specification.....	28
3.4.6 Module Interface Description.	28
3.5 Flame / Fire sensor LM393 Module.	29
3.5.1 Working principle of a flame sensor.	29
3.5.2 Types of Flame sensors.	29
3.5.3 Closed circuit video cameras.	30
3.5.4 Detection range of a flame sensor.....	30
3.5.5 Uses of fire sensors.	31
3.5.6 Module Features and Specifications.....	31
3.5.7 Board Schematic.	32
3.5.8 Module instructions.	32
3.5.9 Pin connections.	32
3.5.10 Electrical characteristics.....	33
3.6. MQ2 Gas Sensor Module.	34
3.6.1 Working Principle.....	34
3.6.2 Why install a smoke alarm in your home?.....	34
3.6.3 Basic Circuit.	35
3.6.4 Specifications.	35
3.6.5 Applications.....	36
3.6.6 Structure and configuration.....	36
3.6.7 Basic test loop.	36
3.6.8 Sensitivity Characteristics.	37
3.6.9 Influence of Temperature/Humidity.....	38
3.7 Wi-Fi module Introduction.....	38
3.7.1 Packaging and Dimension.	39
3.7.2 Schematics Diagram.....	40
3.7.3 Features.	40

3.7.4 Parameters.....	41
3.7.5 Pin Descriptions.	41
3.7.6 Pin Mode.....	41
3.7.7 Receiver Sensitivity.	42
3.7.8 Functional Descriptions.....	42
3.7.9 Memory Organization.....	42
3.7.10 Interfaces.	43
3.7.11 Absolute Maximum Ratings.	44
3.7.12 Recommended Operating Conditions.....	44
3.7.13 Digital Terminal Characteristics.	44
3.7.14 RF Performance.....	44
3.7.15 Power Consumption.	45
3.7.16 Reflow Profile.....	45
3.8 HYDZ Buzzer Module.....	46
3.8.1 Applications of Buzzer.....	47
3.8.2 Buzzer Pin Configuration.....	48
3.8.3 Buzzer Features and Specifications.....	48
3.9 128*64 OLED.....	48
3.9.1 Absolute maximum ratings.	49
3.9.2 Electrical characteristics.....	49
3.9.3 OLED pin connections.	50
3.10 Power Supply LM1117 module. (800mA Low-Dropout Linear Regulator).	51
3.10.1 General Description of LM1117 module.....	51
3.10.2 Features.	52
3.10.3 Applications.....	52
Chapter 4 System flow and main function design	53
4.1 System Flow	53
4.1.1 Overview of system processes.....	53
4.1.2 Data Collection Process.....	54
4.1.3 Cloud Platform Transfer Process.....	55
4.2 Wireless Communication	55
4.3 Temperature and humidity collection	56
4.4 Cloud Platforms.....	56
Chapter 5 System debugging	57
5.1 Hardware debugging.....	57

5.2 Software debugging	58
Conclusion.....	62
References.....	63
ACKNOWLEDGEMENTS.....	65
Publications:.....	66
Academic Publication Highlights:.....	66
Appendix:	67

LIST OF FIGURES

Figure 1 1 lever tumbler lock.....	3
Figure 1 2 Edward Callahan of Baltimore	4
Figure 1 3 Marie Van Brittan Brown, and her the first video monitoring system.....	4
Figure 2 1(a) Block diagram. Figure 2.1 (b) Hardware Implementation	6
Figure 3 1 Parts of a microcontroller.....	10
Figure 3 2 Pin assignment of STM32 MCU	13
Figure 3 3 source of HSE clock AC diagram	15
Figure 3 4 Low speed external clock source AC diagram.....	15
Figure 3 5 STM32 Clock tree.....	16
Figure 3 6 STM32F103C8T6 Block diagram	17
Figure 3 7 Minimum circuit design of STM32 MCU used in project.	20
Figure 3 8 [a] DHT 11 Humidity & Temperature Sensor. Figure 3.8(b) DHT11 External circuit diagram of the temperature and humidity sensor	21
Figure 3 9 Typical Application	23
Figure 3 10 Overall Communication Process.....	23
Figure 3 11 MCU Sends out Start Signal & DHT Responses.	24
Figure 3 12 [a] Data "0" Indication Figure 3.12[b] Data "1" Indication.....	24
Figure 3 13 IR Infrared Obstacle Avoidance Sensor Module.	26
Figure 3 14 working mechanism IR Sensor Module.	26
Figure 3 15 Connections IR Sensor Module.....	27
Figure 3 16 IR sensor Module Interface Description.	29
Figure 3 17 LM393 Module.	29
Figure 3 18 Detection range of a flame sensor.....	30
Figure 3 19 Board Schematic of a flame sensor	32
Figure 3 20 Pin connections (top view)	32
Figure 3 21 MQ2 Gas Sensor Module.....	34
Figure 3 22 Basic Circuit of MQ2 Gas Sensor Module.	35
Figure 3 23 Structure and configuration of MQ2 Gas Sensor Module.....	36
Figure 3 24 Basic test loop of MQ2 Gas Sensor Module.....	36
Figure 3 25 typical sensitivity characteristics of MQ2 Gas Sensor Module.....	38
Figure 3 26 typical temperature and humidity characteristics.	38
Figure 3 27 ESP8266EX Block Diagram	39
Figure 3 28 [a] [Module Pin Counts, 8 pin, 14.3 mm * 24.8 mm * 3.0 mm. Figure 3.28[b] Top View of ESP-01 Wi-Fi Module.....	39
Figure 3 29 Schematics of Esp-01 Wi-Fi Module	40
Figure 3 30 HYDZ Buzzers.....	46
Figure 4 1 Partial system flowchart.....	53
Figure 4 2 Data acquisition flowchart.....	54
Figure 4 3 Cloud platform flowchart	55

Figure 4 4 POST packets	56
Figure 5 1 Physical connection diagram	57
Figure 5 2 Cloud platform generates an operation diagram	58
Figure 5 3 Cloud platform product function addition diagram.....	59
Figure 5 4 Panel Control Layout Diagram	59
Figure 5 5 Added equipment test diagram.....	60
Figure 5 6 Serial port debugging data diagram	61
Figure 5 7 Mobile device data view diagram.....	61

Abbreviations

OLED	organic light-emitting diode
OEM	original equipment manufacturer
LQFP	Low Profile Quad Flat Pack
RISC	Reduced Instruction Set Computer
APB	Advanced Peripheral Bus
RAM	random access memory
USB	Universal Serial Bus
MCU	Microcontroller Unit
HVAC	heating, ventilation, and air conditioning
SRAM	Static random-access memory
PWM	Pulse Width Modulation
PLC	programmable logic controller
POR	Power-On Reset
PVD	programmable voltage detector
DMA	Direct Memory Access
JTAG	Joint Test Action Group
GPIO	general-purpose input/output
HSE	high-speed external clock
GNSS	Global Navigation Satellite System
TTL	Transistor–transistor logic
IPX	Internetwork Packet Exchange
GNSS	Global navigation satellite system
PSTN	public switched telephone network
USART	universal synchronous and asynchronous receiver-transmitter
3GPP	3rd Generation Partnership Project
PCB	printed circuit board
EEPROM	Electrical Erasable Programmable Read-Only Memory
SIM	Subscriber Identity Module
LSE	Low-speed external clock
SFR	Special Function Registers
CAN	Controller Area Network
VLR	visitor location registration
HLR	home location register
PIR	passive infrared sensor
SPI	Serial Peripheral Interface
ADC	Analog to Digital Converter
DSP	digital signal processor
CPU	central processing unit
SMS	short message services
LED	Light Emitting Diode
ROM	Read-Only Memory
PDR	Power Down Reset
RTC	Real-Time Clock
SWD	Serial wire debug
PPS	Pulse per second

Chapter 1 introduction

1.1 Overview

Because of the quick advancement in electronic advances and their joining with customary structure industry, the idea of keen or savvy home automation framework has been received by specialists and way of life experts. The first keen house thought was introduced in the mid 1980's as a task of the National Research Center of the National Association of Home Builders (NAHB) with the collaboration of an assortment of major mechanical accomplices. The essential thought of home computerization is to utilize sensors and control frameworks to screen a home, furthermore, as needed to be changing the different instruments that give heat, ventilation, lighting, and different administrations ^[1]. Present day homes need to utilize numerous machines for agreeable human existence and the interest in energy will increment. For that, shrewd home can accomplish an effective and smart energy the executives and utilization through planning, controlling, analytic, and checking the various apparatuses. Financially, it will increment the long lasting of the machines and lessening the upkeep costs ^[2-5]. Considerable efforts were made to the improvement of faraway manage structures for domestic automation. The in advance work of such structures are specifically primarily based totally on the usage of phones, such as a phone-primarily totally based gadget for domestic automation the use of faraway controlling several of them primarily based totally on a private computer approach ^[6-9]. Early domestic automation commenced with labor-saving machines ^[10].

As residents have real needs for high-quality life in terms of safety, comfort, and convenience, many smart home-related equipment and products have continuously emerged in the market. However, research and analysis of the market status and development prospects of the entire industry, domestic market does not yet have a set of unified standards for smart homes, and its products and equipment also have disadvantages such as high installation and use prices, inconvenient repairs, and high maintenance costs. This also makes consumers choose to use them in pursuit of safety and comfort. When it comes to smart homes, there are certain concerns.

Self-contained electric powered or fuel line powered domestic home equipment have become feasible with inside the 1900s with the creation of electrical energy distribution and caused the creation of washing machines, water heaters, refrigerators, stitching machines, dishwashers, and garments dryers. In 1975, the primary widespread motive domestic automation community technology, X10, turned into developed. It is a communicate protocol for digital devices. It mostly makes use of electric powered energy transmission wiring for signaling and control, in which the alerts contain quick radio frequency bursts of virtual data and stays the maximum broadly available. By 1978, X10 merchandise protected a 16-channel command console, a lamp module, and an equipment module. Soon after got here the wall transfer module and the primary X10 timer. By 2012, with inside the United States, in step with ABI Research, 1.5 million domestic automation structures had been established. Per studies organization Statista extra than forty-five million clever domestic gadgets could be established in U.S. houses with the aid of using the end of the 12 months 2018. The phrase "demotics" (and "demotica" while used as a verb) is a contraction of the Latin phrase for a home and the phrase robotics. The phrase "clever" in "clever home" refers to the machine being privy to the kingdom of its devices that is completed via the statistics and conversation technologies (ICT) protocol and the Internet of Things (LOT). In 2011, Microsoft Research discovered that domestic automation should contain excessive value of ownership, inflexibility of interconnected gadgets, and terrible manageability. When designing and growing a domestic automation system, engineers bear in mind numerous elements which include scalability, how nicely the gadgets may be monitored and controlled, ease of set up and use for the consumer, affordability,

speed, security, and cap potential to diagnose issues. Findings from Internet of things Control confirmed that customers prioritize ease-of-use over technical innovation, and even though customers understand that new linked gadgets have an exceptional cool factor, they are now no longer pretty prepared to apply them of their very own houses yet.

Historically, structures had been bought as whole structures wherein the patron is predicated on one seller for the complete machine together with the hardware, the communications protocol, the crucial hub, and the consumer interface. However, there now are open hardware and open supply software program structures which may be used in preference to or with proprietary hardware. Many of those structures interface with patron electronics consisting of the Arduino or Raspberry Pi, which are without difficulty handy on-line and in maximum electronics stores. In addition, domestic automation gadgets are increasingly interfaced with cell telephones thru Bluetooth, taking into consideration multiplied affordability and customizability for the consumer.

1.2 History of home security systems

In recent years, as home security have begun to provide homeowners with things like keyless entry, home security has gotten rid of simple control panels and deadlocks. Home security has developed into a cool and desirable high-tech product on the wish list of many homeowners. Although the Jetson era has not yet arrived, the technology behind modern smart homes, such as the online viewing function of home security cameras, may be enough to shock our ancestors. A lot has been invented in terms of security. Not just the camera, but also things like these Verify biometric smart safes and many other things have changed and updated from the past as well^[11]

But today's security system is not just the product of technological advancements in the past few years; the foundation of an intelligent security system was laid several generations ago. To understand how home security has evolved into what it is today, you need to look back in time.^[12]

1. Ancient security systems

The history of home security systems began when humans built the first houses, although they were quite different from the security systems we use today. People have an instinct to protect the safety of their homes, families, and property. Perhaps because of the history of war, or because of our inherent responsibility to protect each other, we cultivate this instinct. Since the birth of humankind, this instinct has opened the door to the development of simple and complex home protection methods.

In ancient history, people used fire alarms and watches to protect themselves from enemies and wild animals. These early societies could not install security systems, so they had to devote time and energy to ensuring people's safety. People often take advantage of the landscape to build villages with natural barriers and landmarks to resist the enemy. Even before electricity was discovered and introduced into modern technology, security systems took many forms. With the advancement of technology, humans began to use architecture and design for security purposes. Rulers and monarchs use suspension bridges, moats, ramparts, and small archery holes to protect themselves from invaders and enemy forces. Personal weapons allow people to use offensive measures to defend themselves. Throughout history, humans have always relied on animals and pets to alert them to intruders. Guard dogs are a classic form of animal security systems. For generations, humans have trained dogs to warn their owners of intruders, and even hunt down and attack them to drive them away. To this day, animals remain a means of protecting homes and property.

2. Who invented the first home security system?

What about inventions in modern history? When was the home security system invented and who created the current home security trend? Benjamin Franklin was one of the first to introduce a home security system to the United States in the 18th century. The security system itself is nothing more than a small mirror attached to the windows of the second or third floor of the building. If you visit Philadelphia, you can still see this device outside of Philadelphia's historic buildings.

The tilt angle of the mirror allows people in the house to unknowingly see the person standing at the front door. Benjamin Franklin said that when his mother-in-law knocked on the door, he used his safety glasses to know when to sneak out the back door. In all respects, avoiding uninvited guests is a reason to establish a personal home security system.

The mirror is not a very fancy method of home security, but it shows that home security can be simple and effective. Since those humble beginnings in the 1700s, American domestic security has developed rapidly throughout the 19th and 20th centuries. Although this technology is still in its infancy, it has already affected the modern security systems we use in homes and businesses. Without these inventions, today's security systems would be different.

3. Early attempts at home security

The history of security systems is closely related to other important developments such as electromagnets, batteries, and telegraphs. As these new and advanced developments became common, inventors and enthusiasts began testing new methods to improve home security. It just takes a little bit of wisdom to combine existing ideas into new things. These early inventors may not realize how much their ideas about the development of security systems will have in the coming decades or even centuries.

Early 1700s - English inventor Tildesley is credited with the first creation of an intrusion, door alarm by using a set of chimes mechanically linked to the door lock.

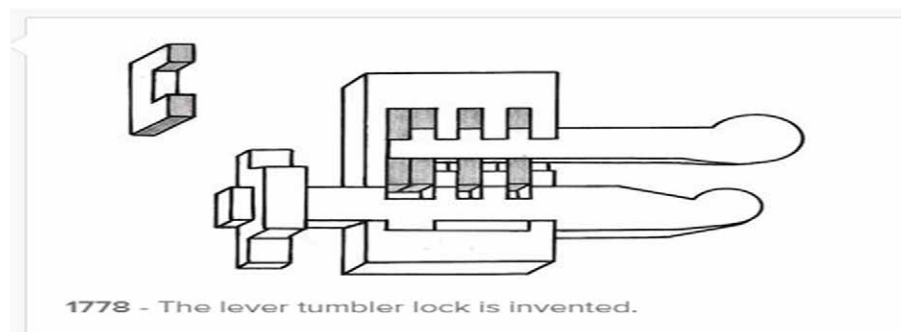


Figure 1 1 lever tumbler lock

In the mid-1800s, a New Hampshire electrician named Moses Farmer used electromagnets to periodically open and close circuits and then ring the bell. The other two later used his design for home security. Boston Secretary August Pope used Farmer's design to ring the bell in response to intruders as part of his electronic home security system. William Channing used the telegraph, the code wheel, and Farmer Bell technology to create a city-wide system to alert the local fire department of an impending fire.

Edward Callahan of Baltimore made great progress in 1875, developing a system that allowed businesses and households to call messengers by using remote phone booths to communicate

with the central location. Through this enhancement, the security system will issue burglar alarms to homes and local law enforcement agencies. It was a popular service that allowed Callahan to sell his invention to investors who founded a new company called the American District Telegraph (now known as ADT).

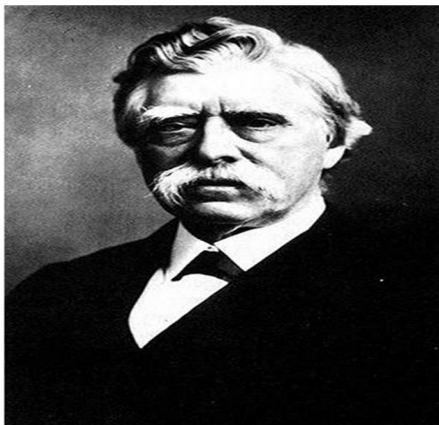


Figure 1 2 Edward Callahan of Baltimore

4. A broader market.

In the late 1900s, enterprising merchants like Edwin Holmes promoted Pope and Channing's designs to a wider market and sold them outside of New England for the first time. Holmes felt that New York was a mature market for his security system, and soon large retailers and homeowners began using his warning system to stop intruders at night. As these developments entered the wider market, others began to innovate and build on these early signaling systems.

5. Security in the 20th century

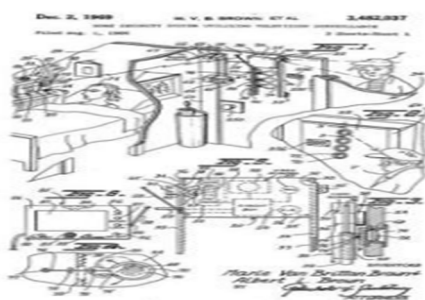


Figure 1 3 Marie Van Brittan Brown, and her the first video monitoring system

Innovation in security has not slowed down in the 20th century. After World War I, people started renting outdoor keys after dark to make sure their houses were locked. In the 1960s, nurse Marie Van Brittan Brown developed the first video surveillance system and the first system that could open doors remotely. Although the image quality of these early security videos is blurry, these trailers herald our current take on home security.

The 1990s brought one of the most important breakthroughs in history: the Internet. Before this, people had never experienced this level of connection and fast data transmission. People can now connect with other people around the world if they have an internet connection. As increased devices are ready to connect to the Internet, people are starting to use them to protect

their homes in a more cautious and convenient way. The combination of video surveillance, remote lockdown, and the Internet paved the way for modern security systems.

6. Today's connected home security systems.

In America today, we can use almost countless forms of home security to meet various needs or lifestyles. You can stay away from home, even on the other side of the world, and still, use your home security features. Things like smart sensors and controllers can give you unprecedented control over your home security.

Professional security companies can install several types of sensors to monitor your home. The sensors can track things like activities in your home, doors opening, broken windows, smoke, carbon monoxide levels, and even the temperature in your home. These sensors are connected to the controller, allowing you to change your home with a powerful smartphone. The sensor sends updates to your phone through a security app, so you can keep track of what is happening at home. You can then use the controller to influence other operations.

If your security system supports home automation, you can use your smartphone as a controller. With their help, you can adjust the temperature of your home and take advantage of features that can deter intruders and burglars. If you are not at home and your security app is warning you that someone is trying to break in, you can remotely turn on the lights to make it appear that someone is awake, and you know you are there.

You can even lock the door and close the garage door after the app tells you forgot to lock the door when you leave work. Returning to the open garage door and the anxiety that comes with it may be outdated.

Some modern security systems allow you to speak on the phone so that anyone at your door can hear you. If you see an intruder, you can tell them that you have called the police, which may be enough to scare them. You can use this function to thank the postal worker for leaving the package or say hello to a friend who just passed by.

1.3 Research significance of the project.

The design has carried out research on the intelligent home system. The main work content includes circuit schematic drawing, cloud platform construction, mobile software writing and hardware circuit design, which can realize the temperature and humidity, light intensity, indoor gas concentration data of the home environment for monitoring and alarming, identity can be verified by collecting fingerprint information, and visitor information can be viewed by logging in to the mobile terminal account. The various functions of the system can be monitored and controlled on the wireless receiving device. The remote-control function realized through the cloud platform also enables users to have a real-time and accurate grasp of the home environment through the smart home system even when the user is away from home, showing that users pay the most attention to convenience and safety in many aspects.

Chapter 2 the Design of Hardware.

2.1 Introduction of the system.

This project circuit consists of an STM32 microcontroller which is physically connected with four sensors. IR sensor, smoke detection, flame detection, temperature, and humidity sensor. The microcontroller takes the data from the sensors and uploads it to the cloud plat through wireless communication. When the microcontroller detects any critical value mentioned in the code program, it will send a control signal to activate the alarm, also it will ON the ventilation system and door through relay switches, and the OLED panel will display the data of the sensors and circuit functionality processes.

Relays are used to protect the electrical system and to minimize the damage to the equipment connected in the system due to over currents/voltages or some other abnormal condition that occurred in the system. In this project, the relay is used as a switch, when the relevant condition meets it will switch ON the ventilation equipment and auto-lock door lock system.

The block diagram of the system is shown in Figure. 2.1(a). The design has carried out researched on the intelligent home system. The main work content includes circuit schematic drawing, cloud platform construction, mobile software writing, and hardware circuit design, which can calculate the temperature and humidity, light intensity, indoor gas concentration data of the home environment for monitoring and alarming, identity can be verified by the visitor information and viewed by logging in to the mobile terminal account. The various functions of the system can be monitored and controlled on the wireless receiving device. The remote-control function realized through the cloud platform also enables users to have a real-time and accurate grasp of the home environment through the smart home system even when the user is away from home, showing that users pay the most attention to convenience and safety in many aspects. The hardware implementation of the project is shown in Figure. 2.1(b).

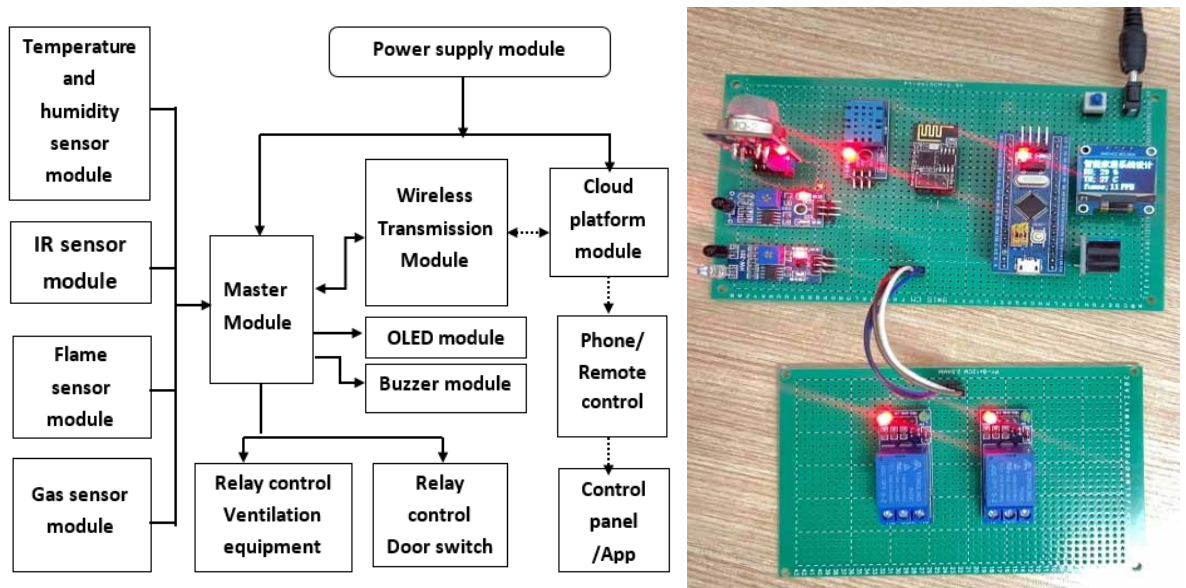


Figure 2.1(a) Block diagram. Figure 2.1(b) Hardware Implementation

Software design: First, it is the construction of the cloud platform and the design of the display operation interface displayed on the mobile terminal or webpage used by the user. Second, the design of the driver for temperature and humidity sensors, smoke sensors, access control recognition, etc. And the temperature and humidity sensor in the information data collected in the actual home environment is packaged and sent to the various interfaces of the user for display and provides a data source for the next control design. The third is to design a smoke drive and alarm drive. The last is to design the integration program.

2.2 STM32F103 Microcontroller.

STM32F103 microcontroller is manufactured by STMicroelectronics with ARM Cortex-M3 architecture. STM32F103 microcontroller comes with forty-eight pin LQFP (Low Profile Quad Flat Pack) package. It includes a 72MHz operating frequency, fast embedded memories, a wide range of enhanced I/Os, and a high-performance RISC core with peripherals connected to two APB buses. A comprehensive set of power-saving methods can be used in the design of low-power applications. It has 64 Kb or 128 Kb flash memories and 20 Kb RAM. It supports 4 to 16 MHz frequencies operations. In addition, it has motor control peripherals, CAN, and USB full-speed interface. There are multiple package types available for embedded applications. The MCU STM32 architecture has an easy-to-use principle and will work in applications including motor drives, PC (Personal computer) and gaming, HVAC (heating, ventilation, and air conditioning), and industrial applications. The devices are pin-to-pin software compatible. It has three sixteen bits timers and up to 9 communication interfaces including I²C and three USARTs interfaces. It has eighty input/output ports. The operating dc voltage is from 2.0 to 3.6 volts. The operating temperature ranges are -40 to +85 °C or -40 to +105 °C.

2.3 Temperature and humidity sensor module.

Humidity (or humidity) sensors feel, measure, and report air humidity and temperature. The ratio of humidity and the maximum humidity in the air at a certain temperature is referred to as relative humidity. Relative humidity becomes a crucial factor in the search for comfort.

Moisture is the water in the air. The amount of water in the air will not only affect personal comfort but also affects various manufacturing processes in industrial applications.

For example, in the semiconductor industry, the humidity level or humidity must be properly controlled to ensure proper dilution handling. Humidity control is important for incubators, respiratory equipment, sterilizers, and biological products. Furthermore, the presence of water vapor can also affect various other chemicals, biological and physical processes.

Due to the higher humidity, the environment in the environment can be critical, especially since it seems high. In industry, humidity measurements are often important as they will affect the health and safety of personnel and the cost of the product. As a result, temperature and humidity sensors are especially important.

Humidity measurements are also a crucial factor in time because the presence of humidity indicates the possibility of dew, fog, or precipitation. Higher relative humidity reduces the efficiency of sweat to cool down. This is because sweat evaporation is prevented. The effect is measured in the hot meter. As a result, it often feels hot in the summer, there is a region with higher relative humidity.^[13]

2.4 IR Sensor module.

Many technological innovations change the world. One of them is infrared technology. Infrared technology currently has a wide range of wireless applications, used for object detection and remote control.

The infrared sensor obstacle avoidance sensor board is a cheap obstacle detection solution suitable for robots, smart cars, home security systems, and other electronic applications.

The infrared obstacle sensor module has a built-in infrared transmitter and receiver, which emits infrared energy and searches for reflected infrared energy to detect any obstacles in front of the sensor module. The module has a built-in potentiometer, allowing users to adjust the detection range. Even in natural light or in complete darkness, the sensor has a particularly good and stable response.

Obstacle avoidance sensors are of two types: with 3 and 4 contacts.^[14]

2.5 Flame sensor module.

A fire sensor/flame sensor, as the name recommends, is utilized for distinguishing and reacting to the presence of a fire or fire. The fire sensor typically reacts as solid alerts, fuel line deactivations, (for example, a petroleum gas line or a propane line), and enactment of various sorts of fire concealment frameworks. Fire sensors are used to check whether the heaters are working appropriately. These sensors are likewise utilized in a start framework to make precise moves or to tell the administrator. Fire sensors or fire locators are similarly more exact and swifter than smoke or warmth sensors^[15]

2.6 Gas sensor module.

A commonplace human nose has four hundred sorts of aroma receptors empowering us to smell around 1 trillion distinct scents. Yet at the same time a large number of us do not have the ability to distinguish the sort or convergence of gas present in our climate. This is the place where Sensors comes in. A gas sensor recognizes the presence or centralization of gases in the climate. Considering the centralization of the gas the sensor creates a comparing expected contrast by changing the obstruction of the material inside the sensor, which can be estimated as result voltage. Considering this voltage esteem the sort and grouping of the gas can be assessed. The sort of gas the sensor could recognize relies upon the detecting material present inside the sensor.^[16]

2.7 Buzzer module.

Buzzer is a device that converts audio model into signal. It is used to prompt or alarm. According to unique design and application, it can produce music sound, buzzer, alarm sound, electric bell, and other different sounds. Typical applications include siren, alarm device, fire alarm, air defense alarm, burglar alarm, timer etc. It is widely used in household appliances, alarm system, automatic production line, low voltage electrical equipment, electronic toys, game machines and other products and industries^[17]

2.8 Wireless transmission module.

The wireless transmission module used in this design is ESP8266 the module. There are two ways this wireless module works. First way is the wireless access mode, which connects the module to other networks as a network site to transmit data to other devices. Second way is the client mode, which transmits data information through the module to other wireless networks, and the wireless-client mode also can act as both the sender of the data and the network to receive the data. In this design the wireless and client mode, that is, the AP-STA mode, which can be used as a client or server and can also be used as a commonly used Bluetooth connection module, in the configuration of IP address and port number, and other devices successfully docking, the two sides can transfer data to each other.

2.9 Power supply module.

A power source is an electrical device that provides power to an electrical load. The main function of the power supply is to convert the current from the power supply into the correct voltage, current and frequency to supply power to the load. Most power supplies will take power from AC mains and convert it into a usable DC for use in electronic gadgets. During this cycle, a power supply is filling various jobs, including Transforming AC from a mains supply into a consistent DC. Keeping any AC from meddling with a DC supply yield.

Chapter 3 the Design of Hardware Module

The system consists of STM32 Micro-controller, Temperature and humidity sensor module, IR sensor Module, Flame sensor module, Gas sensor module, Buzzer module, Wireless Transmission Module, OLED module, two relays, and Cloud platform module. The microcontroller takes the data from the sensors and uploads it to the cloud plat through wireless communication. When the microcontroller detects any critical value mentioned in the code program, it will send a control signal to activate the alarm, also it will ON the ventilation system and door through relay switches, and the OLED panel will display the data of the sensors and circuit functionality processes.

3.1 Introduction to microcontroller

A micro-controller is additionally known as associate embedded controller. it is a little computer on one metal compound semiconductor computer circuit chip that contains one or additional processors with memory and programmable input/output peripherals. The integrated circuit chip sometimes includes program memory within the ferroelectric RAM, NOR non-volatile storage or OTP store and a small quantity of RAM. On the contrary, micro-controllers are designed for embedded applications. The applications contain numerous separate chips employed in micro-processors or different general computers. Micro-controllers are used for automatic control product and equipment, like automobile engine control systems, remote controls, ready-made medical equipment, workplace machines, power tools, electrical appliances, toys, and different embedded systems.

By reducing the dimensions and value of the design, employing a separate micro-processor, memory, and input/output devices, micro-controllers produce it is economical to digitally manage further devices and processes. Mixed-signal micro-controllers are common and contain essential parts for dominant non-digital electronic systems. Within the context of the net of Things, the micro-controller is an economical, common suggests that of collecting, sensing, and exploitation the physical world as a contemporary tool. a number of the micro-controllers use four-bit words and operate at low frequencies up to four kilohertz to utilize low power (single-digit milli watts or microwatts). They will sometimes maintain functions whereas waiting for the program, like pressing a button or alternative interrupts. The facility consumption throughout sleep (the processor clock and most peripherals are off) will solely be Nano volts that are terribly appropriate for long-lived battery applications. Alternative micro-controllers would play a key role in performance because of they are going to like in operation as digital signal processor (DSP) like clock signals and low power consumption. A microcontroller can seem like a tiny low computer, which is due to the essential elements are

within it, as well as CPU, RAM, bus interference, Flash memory, audio/output ports, and in several cases, the Electrical eradicable Programmable computer memory (EEPROM).

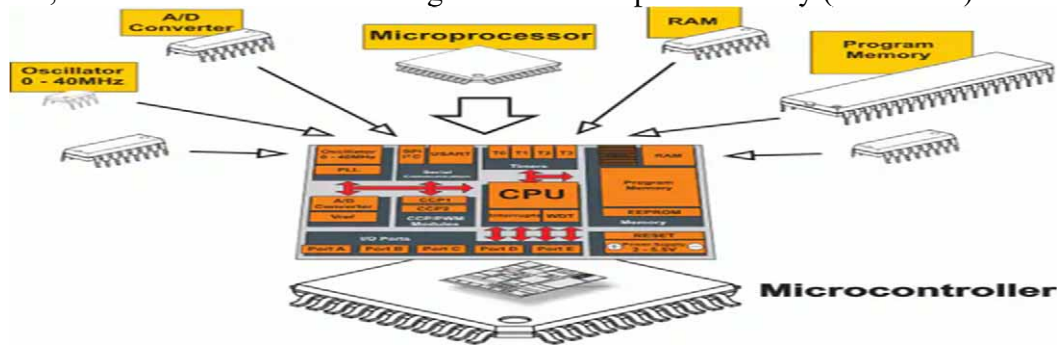


Figure 3 1 Parts of a microcontroller

1. Microprocessor (CPU)

A microprocessor is named a process unit [CPU] C.P.U. | central processor | processor | mainframe | electronic equipment | hardware | computer hardware } device, which is employed to hold and rewrite data to perform the appointed task efficiently. Through the central processing unit, all microcontroller parts are connected to a specific system. Directions obtained through programmable memory may be decoded by the CPU.

2. Memory

In a microcontroller, the chip acts as a microprocessor because of it stores all data } and programs. The microcontrollers are designed with a couple of RAM / ROM / flash reminiscences to store supply code.

3. Input/output ports

Input/output ports are accustomed connect the micro-controller with real-world applications. Inputs receive information changes in the real world, like temperature sensing, pushbuttons, motion sensing and lots of more. The input info then goes to the mainframe and decides what to try and do with input information. to try and do an exact command supported a certain price from the input, information signal send to the output ports, wherever output devices connected with ports adore LED lights, motors, and many more.

4. Serial ports

Serial ports are accustomed give serial interfaces between the micro-controller and numerous different peripherals like parallel ports.

5. Timers

A micro-controller is also in-built with timers or counters. The timers and counters are used to management all the operations of temporal order and counting. the most work of the counter is to count external pulses whereas the operations performed by timers are clock functions, pulse generations, activity frequency, modulations, creating oscillations, etc.

6. Analog to digital converter (ADC)

Analog to Digital device is utilized to convert analog signals to digital ones. For Analog to Digital Converter, the input signals are Analog, and the production of a digital signal is used in numerous digital applications like measurement gadgets.

7. Digital to analog converter (DAC)

Digital to analog converter executes opposite functions that to Analog to digital converter. This device is used to manage analog appliances like AC motors, etc.

8. Interpret control.

This controller is employed for giving delayed control to a running program. The interpretation is either internal or external.

9. Special functioning block

Some special micro-controllers factory-made for special devices like house systems, robots, etc., embody a special perform block. This special block has some further ports to hold out some special operations.

These days there are numerous classes of micro-controllers in line with their architecture, memory, bits, and instruction sets, samples of these micro-controllers are 8051 micro-controller, PIC microcontroller, AVR microcontroller, ARM micro-controller. The controller I exploit for my style is ARM Cortex-M3 STM32F103C8T6 micro-controller.

3.2 ARM Cortex-M3 STM32F103 micro-controller

STM32F103 Medium Density Performance Series family includes seventy-two rate frequency superior ARM Cortex provide} 32-bit reduced instruction set computer core, high speed embedded memory (up to 128 Kbyte non-volatile storage and twenty Kbyte SRAM), and a large vary of extended Inputs/Outputs and peripherals connected to 2 APB buses. Every device offers two 12-bit ADCs, three common 16-bit timers, and a PWM timer, moreover as normal and advanced communication interfaces-up to two I2Cs and SPIs, one USB, one CAN, and three USARTs. The device operates on an influence supply of 2.0 to 3.6 V. they are offered in each temperature ranges of -40 to + eighty-five ° C and between -40 to one zero five ° C. The power-saving model technique permits for the creation of low-power applications.

The STM32F103 Medium Density Performance Series family includes six differing types of devices, from thirty-six pins to a hundred pins. Reckoning on the device selected, different sets of peripherals are included; the subsequent description provides a summary of all peripherals during this family.

These options build the STM32F103 medium-density performance series microcontroller family appropriate for a large sort of applications, admire application control, motor, hand-held equipment, and medical, laptop and game peripherals, industrial applications, GPS platforms, inverters, PLCs, printers, alarm system, scanners, video intercoms, and HVACs.^[18]

3.2.1. All features

- ARM 32-bit Cortex M3 Central Processing Unit Core
 - The maximum Operation frequency range is 72 MHz, 1.25 DMIPS/MHz performance at zero wait state memory access.
 - Have Single-cycle multiplication and hardware division.
- Memories
 - 64 or 128 Kbytes of embedded Flash memory are available for storing data.
 - 20 Kbytes of embedded Static Random-Access Memory accessed (read/write) at central processing unit clock speed with zero wait states.
- Reset, clock, and power supply management.
 - The operation voltage range is 2V to 3.6 V for application supply and Input/Outputs.
 - Power-On Reset (POR), Power Down Reset (PDR), and programmable voltage detector (PVD)

- 4 MHz up to 16 MHz crystal oscillator
- Internal 8 MHz factory trimmed RC. Internal 40 kHz RC
- Phase-locked loops (PLL) for CPU clock
- 32 kHz oscillator for Real-Time Clock (RTC) with calibration
- Low power
 - Standby, Stop, Sleep, and modes
 - VBAT supply for Real-Time Clock (RTC) and backup registers
- 2x 12-bit, 1 μ s A/D converters (up to sixteen channels)
 - Conversion ranges from 0V to 3.6 V
 - Holding capability and dual sample
 - The temperature sensor
- Direct Memory Access (DMA)
 - 7-channel Direct Memory Access (DMA) controller
 - Peripherals supported, timers, Analogue to Digital C (ADC), SPIs, I²Cs, and USARTs
- Up to 80 fast Input/output ports
 - 26/37/51/80 I/O, all mappable on 16 external interrupt vectors and all 5 V-tolerant
- Debug mode
 - Joint Test Action Group (JTAG) interfaces and Serial wire debug (SWD)
- Seven timers
 - Three 16-bit timers, each with up to four input capture (IC) modes/Output Compare (OC) modes/Pulse Width Modulation (PWM) or pulse counter and quadrature (incremental) encoder input.
 - 16-bit, motor control Pulse Width Modulation (PWM) timer with dead-time generation and emergency stop.
 - two watchdog timers (Independent and Window)
 - SysTick timer 24-bit down-counter
- Up to Nine communication interfaces
 - Up to 2 x I²C interfaces (SMBus/PMBus)
 - Up to 3 USARTs (ISO 7816 interface, LIN, IrDA capability, modem control)
 - Up to 2 SPIs (18 Mbit/s)
 - Controller Area Network (CAN) interface (2.0B Active). with USB 2.0 full-speed interface.^[19]

STM32F103C8T6 has thirty-seven GPIO pins have unfold across four ports “A, B, C, D.” A and B ports have sixteen pins, and C port have three pins, and D port have 2 pins. All pin ports have a current sink/source capability of 6mA. Pull-up and pull-down resistors may be enabled on each pin. Most of the pins have some further practicality likewise as serial ports that receive and transmit information via the UART protocol, the I2C port' functionality of two-wire communication and via the I2C protocol, Serial peripheral interface (SPI) used for serial communication, Pulse breadth Modulation (PWM) and Pin thirteen contains an inherent LED.

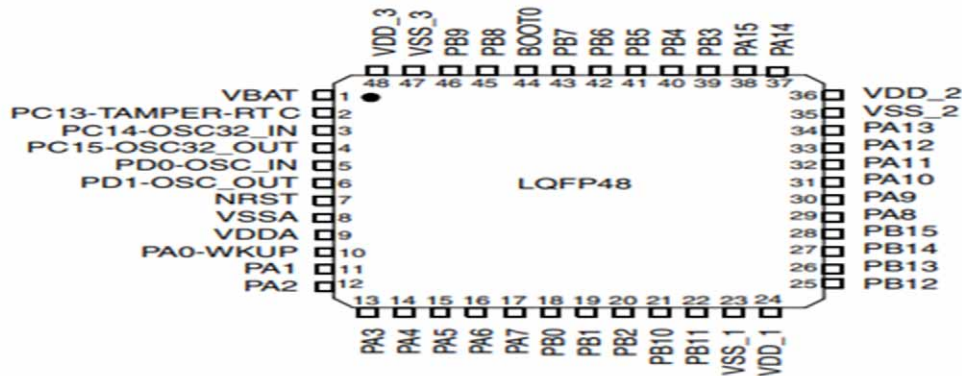


Figure 3 2 Pin assignment of STM32 MCU

Each port has associate input buffer, output driver and has a special quite registers. The output drivers of Ports zero and two, and the input buffers of port 0, may be used for external memory. Port A outputs an occasional computer memory unit from the external memory address and the time-multiplexed by the byte written or read. Port B outputs high bytes from an external memory address once it is sixteen bits wide. Otherwise, the Port B pins still emit the P2 (Port) Special perform Registers content (SFR). All the pins in ports are sixteen bits wide. If the corresponding bit latch within the port Special perform Registers contents (SFR), and a one alternate function are activated. Otherwise, the port pin remains at 0. Its feature is easy and covers a range of tasks.

3.2.2 Clock circuit

An introduction to the concept of your time in Micro-Controller Unit (MCU) is intended by clock circuit in it. The prime operate of the clock circuit half is to regulate the total system design, the running chain of the whole design, and the period of every sentence. Regardless of what quite control chip it is, there will be a clock circuit. If the MCU is that the brain of the entire system design, the clock circuit within the MCU is the heart that provides vital energy to the brain. It endlessly controls the rhythm throughout the employment of micro-controller. It would be hypothetic to assume that if the directions of every microcontroller were not met at identical time, this could confuse declaring the entire system. Within the running place, the quick running will directly create the delay operate invalid, then the entire system will not in sequent execution; we are going to eventually crash and not run. The thought of the concept jump is like people' horns. As so much because the current understanding is concerned, there is a logical confusion in the internal program of the MCU. The inner work address is ready to start with the program or at the address wherever the microcontroller has simply started working. STM32 series MCU chip having reset circuit and clock circuit is like inserting the brain and brain stem beneath human skin. The MCU chip and reset circuit and clock circuit structure the littlest system of the STM32 series MCU.

The clock circuit of the STM32 series is completely different from the normally used STC89 series microcontroller. The clock system is additional complicated however more correct and effective. Completely different from the STC89C52 series Micro-Controller Unit within the skilled category, the STM32 could be a more high-end/STM32 series Micro-Controller Unit. To achieve low power consumption, an additional complicated clock system is intended to realize low power consumption and is termed the clock tree. The clock, which permits peripheral functions, is self-configurable. Since there are heaps of STM32 peripherals and the peripherals employed in completely different comes are jagged, a controlled clock will cut back the facility consumption of the product. Before victimization all peripherals, they need to be clocked.

There are four clock sources for the STM32 micro-controller.

1. High-speed external clock (HSE) External clock.

Used as an external clock source. In the circuit, the crystal frequency can be used in the range from 4 to 16 MHz, but in thirty-two series single-chip microcomputers, 8 MHz crystal oscillators are commonly used.

High speed external clock generated from a crystal. To provide an external high-speed clock the Oscillators can be used ranging from 4 to 16 MHz resonators. All these useful details provided in this paragraph are based on the consequences of characterization obtained using typical external components Throughout this method, to achieve a reduction in signal distortion and smoothing of the startup period is by placing the load capacitor and resonator as close as possible.

Table 3.1 HSE Oscillator characteristics (HSE)=4-16MHz

Parameter	Condition	Min	Type	Max	Unit
Oscillator frequency	NIL	4	8	16	MHz
Feedback resistor	NIL	NIL	200	NIL	K Ω
Recommended load capacitance	Rs=30 Ω	NIL	300	NIL	Pf
HSE driving current	Vdd=3.3V, Vin=Vss with 30Pf load	NIL	NIL	1	mA
Oscillator trans conductance	Start up	25	NIL	NIL	Ma/V
Startup time	Vdd is stabilized	NIL	2	NIL	ms

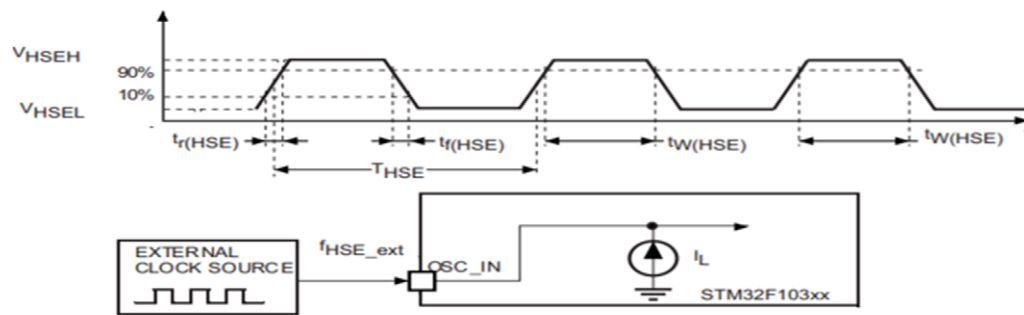


Figure 3.3 source of HSE clock AC diagram

2. Low-speed external clock (LSE)

It is also used as an external clock source. The normal operation is to provide a real-time clock module to the MCU. Therefore, this part of the MCU with the crystal frequency is 32.768 KHz.

Low speed external clock generated from a crystal. With a 32.768 kHz crystal resonator oscillator the low-speed external (LSE) clock can be provided. All the information materials given in this passage are based on characterization consequences obtained with typical external components specified in the table. For this purpose, the resonator and the load capacitors must be mounted near the oscillator pins to minimize output distortion and startup stabilization time.

Table 3.2 LSE Oscillator characteristics ($f_{LSE} = 32.768$ kHz)

Parameter	Condition	Min	Type	Max	Unit
Feedback resistor	NIL	NIL	5	NIL	M Ω
Recommended load capacitance	Rs=30K Ω	NIL	NIL	15	Pf
LSE driving current	Vdd=3.3V, Vin=Vss	NIL	NIL	1.4	mA
Oscillator trans conductance	NIL	5	NIL	NIL	μ A/V

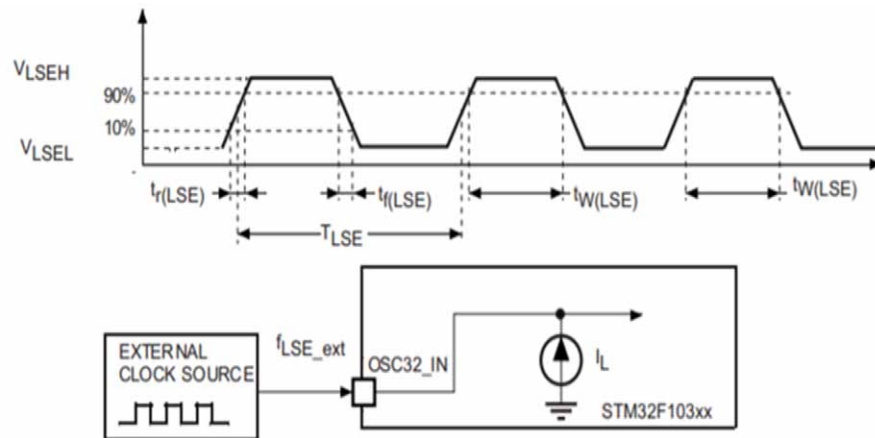


Figure 3 4 Low speed external clock source AC diagram

3. High-speed internal clock (HSI) Internal clock.

An internal clock refers to the internal clock circuit of the MCU. It produces an oscillation frequency from the internal RC oscillator. Usually, the frequency produced is 8 MHz, but since it is a clock-like circuit, the resulting oscillation frequency is not good than the peripheral crystal oscillator, so it is difficult to use a high-speed internal clock accurately.

4. Low-speed internal clock (LSI).

An enclosed RC oscillator additionally produces an oscillation frequency. Its objective is to produce the time clock module to the micro-controller, and the frequency may be about 40KHz. its constant internal clock, which is a clock-like circuit. The resultant oscillation frequency does not be constant because the peripheral crystal oscillator. So, the employment of the low-speed internal clock is additionally unsettled

The choice of employing a high-speed external clock is also thanks to the external clock is additional stable and precise, and the most clock frequency is issued to the SYSCLK by multiplying, and the optimum performance of the central processing unit is formed use of. The peripheral should be enabled to activate the connected clock; otherwise, the STM32 will not be in work properly.

If a High-speed internal clock is employed as a phase-locked loop with an input of 64MHz, then we can succeed the absolute best system clock frequency. to verify the supply of USB, perform all the time then the High-speed external clock and the phase-locked loop ought to be aligned with the USB clock running at forty-eight Mc to meet an Analog to digital conversion time of 1 μ s, Advanced Peripheral Bus two should be at fourteen MHz, twenty-eight MHz, or fifty-six MHz

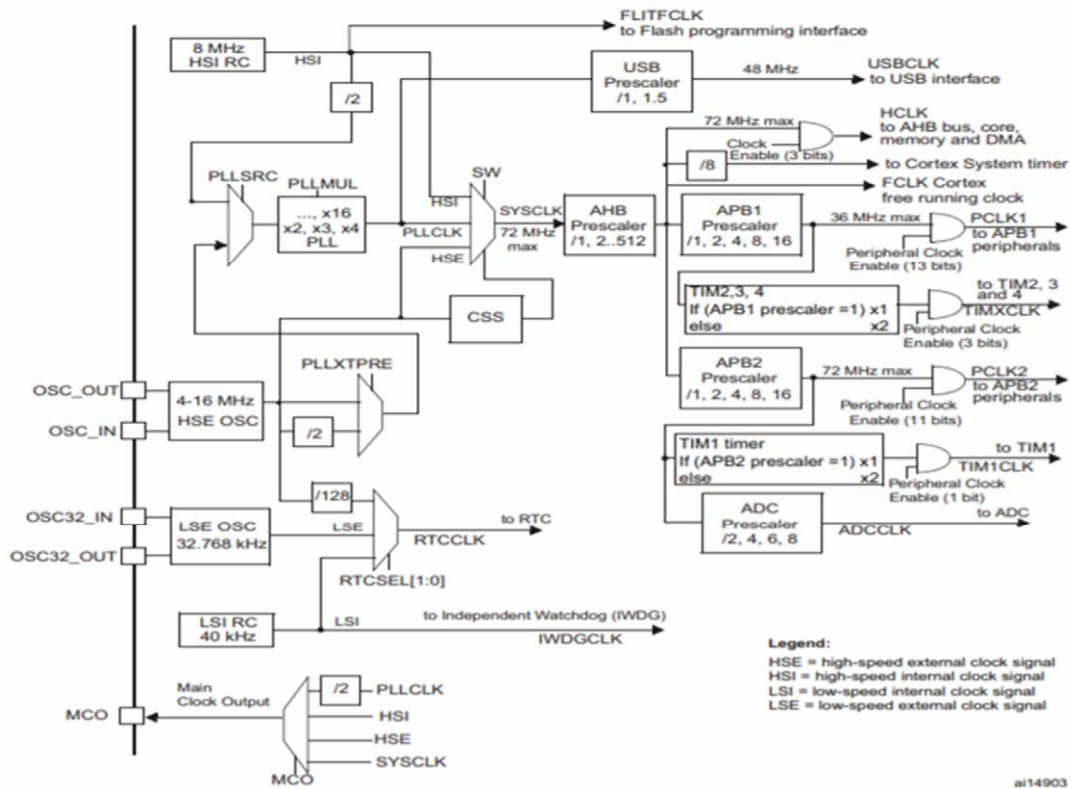


Figure 3 5 STM32 Clock tree

3.2.3 Reset circuit.

In each micro-controller, there will be a reset circuit. If the clock circuit represents the heart, then the reset circuit haunts a valued a part of the brain. The reset circuit of the MCU is that the same because the restart circuit of our personal computer. Also, a similar as the time knowledge refresh once we think about it. Once our computer is in use and it crashes, we press the restart button to let the pc begin from the start address. This is often the same for the micro-controller unit. Once the Micro-controller unit program is within the method of operation, it is interfered by some logic error or by some external setting interference. The running program that is within the method of operation can run away, once we have a tendency to push the reset button, the within the program will mechanically execute from the start.

The prime style of the reset circuit is to bring the system to its initial settings. The reset circuit of the microcontroller we used is principally divided into 3 sorts reset mode the power-on reset, system reset, and backup space reset.

3.2.4 MCU block diagram.

The ARM Cortex-M3 PROCESSOR obtains a state of an art family possessing all the vital tools required for its normal operating. Confining opinion of economical working and execution that address gift technology trend with low price and restraining its power and pin counts together with providing wonderful method likewise as steady response to external interrupts.

The ARM Cortex-M3 thirty two-bit RICS processor options outstanding code potency which ends within the outstanding execution of the given program that may solely be hoped from an ARM processor. The dimension of memory unremarkably out there is with 32 and 64-bit devices.

Therefore, all the devices that are available STM32F103xxx containing AN embedded ARM core are operational with all ARM devices and applications.

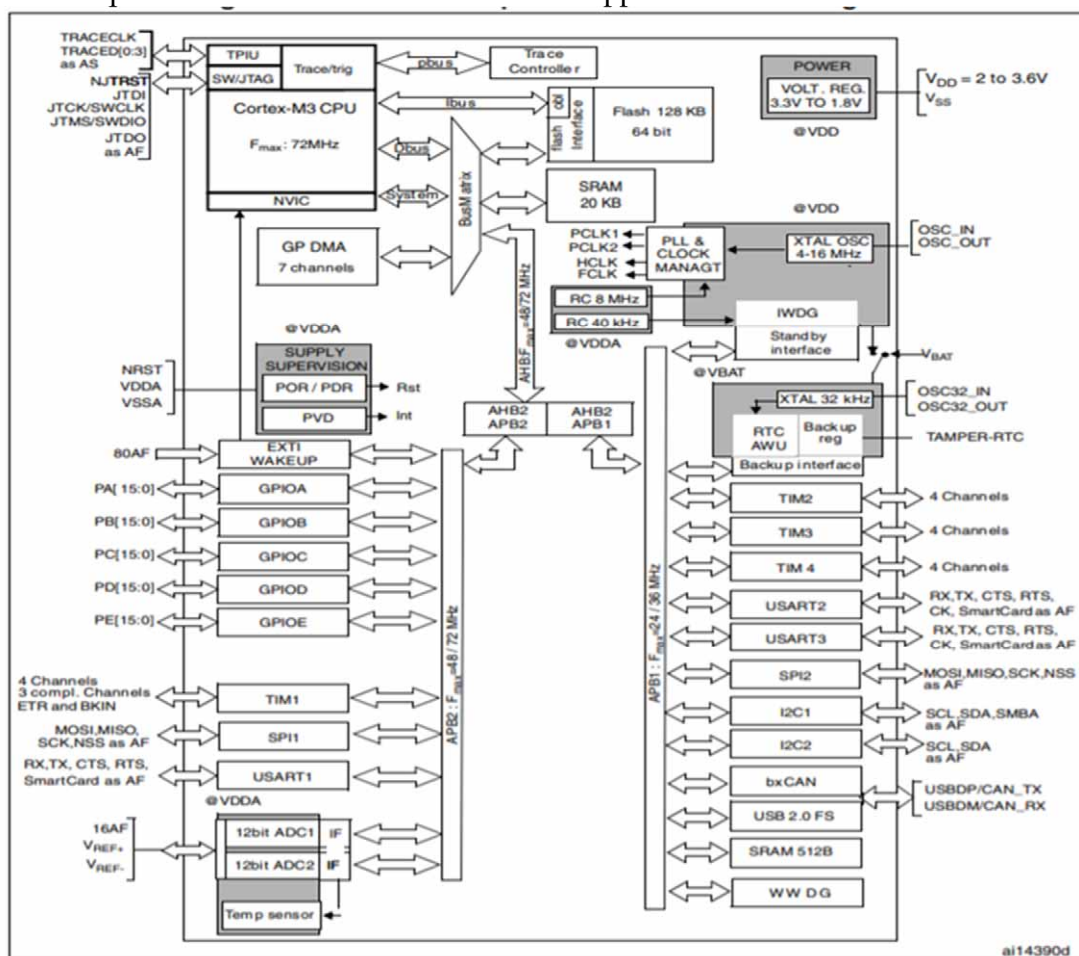


Figure 3-6 STM32F103C8T6 Block diagram

3.2.5 Pin description of STM32F103C8T6

Table 3.3 STM32 Pin description

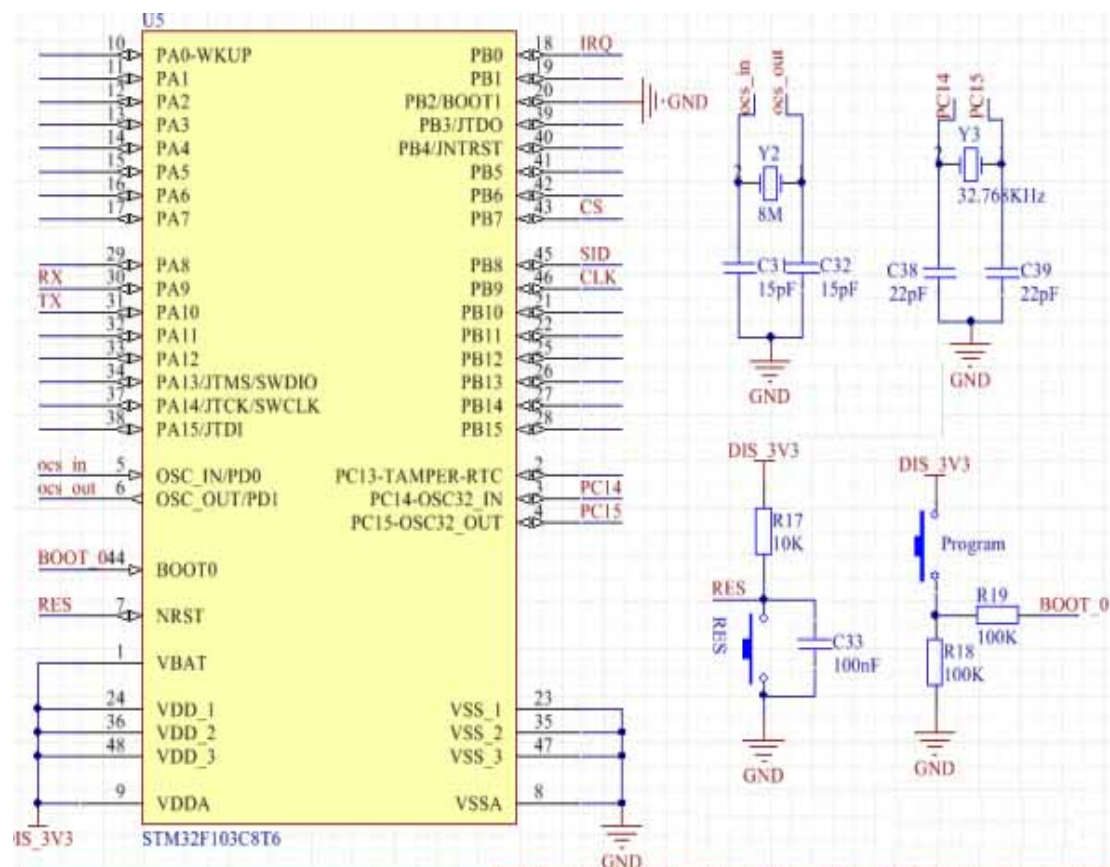
Pin#	Pin Name	Type	I/O Level	Main Function	Alternate function	
					Default	Remap
1	Vbat	S	NIL	Vbat	NIL	NIL
2	PC13-TAMPER-RT C	I/O	NIL	PC13	TAMPER-RTC	NIL
3	PC14-OSC32_IN	I/O	NIL	PC14	OSC32_IN	NIL
4	PC15-OSC32_OUT	I/O	NIL	PC15	OSC32_OUT	NIL
5	OSC_IN	I	NIL	OSC_IN	NIL	PD0
6	OSC_OUT	O	NIL	OSC_OUT	NIL	PD1
7	NRST	I/O	NIL	NRST	NIL	NIL

8	Vssa	S	NIL	Vssa	NIL	NIL
9	Vdda	I/O	NIL	Vdda	NIL	NIL
10	PA0-WKUP	I/O	NIL	WKUP/USART2_CTS/ADC12_IN0/TIM2-RT2-CT	WKUP/USART2_CTS/ADC12_IN0/TIM2-CH1-ETR	NIL
11	PA1	I/O	NIL	PA1	USART2_RTS/ADC12_IN1/TIM2-CH2	NIL
12	PA2	I/O	NIL	PA2	USART2_TX/ADC12_IN2/TIM2-CH3	NIL
13	PA3	I/O	NIL	PA3	USART2_RX/ADC12_IN3/TIM2-CH4	NIL
14	PA4	I/O	NIL	PA4	SPI1_NSS/USART2_CK/ADC12_IN4	NIL
15	PA5	I/O	NIL	PA5	SPI1_SCK/ADC12_IN5	NIL
16	PA6	I/O	NIL	PA6	SPI1_MISO/ADC12_IN6/TIM3_CH1	TIM1_BKIN
17	PA7	I/O	NIL	PA7	SPI1_MOSI/ADC12_IN7/TIM3_CH2	TIM1_CH1N
18	PB0	I/O	NIL	PB0	ADC12_IN8/TIM3_CH3	TIM1_CH2N
19	PB1	I/O	NIL	PB1	ADC12_IN9/TIM3_CH4	TIM1_CH3N
20	PB2	I/O	FT	PB2/BOOT1	NIL	NIL
21	PB10	I/O	FT	PB10	I2C2_SCL/USART3_TX	TIM2_CH3
22	PB11	IO	FT	PB11	I2C2_SDA/USART3_RX	TIM2_CH4
23	Vss1	S	NIL	Vss1	NIL	NIL
24	Vdd1	S	NIL	Vdd1	NIL	
25	PB12	I/O	FT	PB12	SPI2_NSS/I2CS_SMBAL/USART3_CK/TIM_BKIN	NIL
26	PB13	I/O	FT	PB13	SPI2_SCK/USART3_CTS/TIM1_CH1N	NIL
27	PB14	I/O	FT	PB14	SPI2_MISO/USART3_RTS/TIM1_CH2N	NIL
28	PB15	I/O	FT	PB15	SPI2_MOSI/TIM1_CH3N	NIL
29	PA8	I/O	FT	PA8	USART1_CK/TIM1_CH1/MCO	NIL
30	PA9	I/O	FT	PA9	USART1_TX/TIM1_CH2	NIL
31	PA10	I/O	FT	PA10	USART1_RX/TIM_CH3	NIL
32	PA11	I/O	FT	PA11	USART1_CTS/CANRX/USBDM/TIM1_CH4	NIL
33	PA12	I/O	FT	PA12	USART1_RTS/CANTX/USBDM/TIM1_ETR	NIL
34	PA13	I/O	FT	JTMS/SWDIO	NIL	PA13
35	Vss2	S	NIL	Vss2	NIL	NIL
36	Vdd2	S	NIL	Vdd2	NIL	NIL
37	PA14	I/O	FT	JTCK/SWCLK	NIL	PA14
38	PA15	I/O	FT	JTD1	NIL	TIM2_CH1_ETR/PB15/SPI1_NSS
39	PB3	I/O	FT	JTD0	NIL	TIM2_CH2/PB

9						3 TRACESWOS PI1_SCK
4 0	PB4	I/O	FT	JNTRST	NIL	TIM3_CH1/PB 4/SPI1_MISO
4 1	PB5	I/O	FT	PB5	I2C1_SMBAI	TIM3_CH2/SP I1_MOSI
4 2	PB6	I/O	FT	PB6	I2C1_SCL/TIM4_CH1	USART1_TX
4 3	PB7	I/O	FT	PB7	I2C1_SDA/TIM4_CH2	USART1_RX
4 4	BOOT0	I	NIL	BOOT0	NIL	NIL
4 5	PB8	I/O	FT	PB8	TIM4_CH3	I2C1_SCL/CA NRX
4 6	PB9	I/O	FT	PB9	TIM4_CH4	I2C1_SPA/CA NTX
4 7	Vss3	S	NIL	Vss3	NIL	NIL
4 8	Vdd3	S	NIL	Vdd3	NIL	NIL

1. O=Output, I=Input, S=Supply
2. FT=5V Tolerant
3. Function availability depends on the device chosen. For devices with a reduced number of peripherals, the number of peripherals devices used is usually smaller. If a computer has only one SPI and two USARTs, they are called SPI1, USART 1, and USART 2 severally.
4. If several peripherals share the same Input/output pin, then to avoid conflict between these alternate functions at a time only one peripheral should be enabled through the peripheral clock enable bit.
5. The main function will be functional after the switching power up and will rely upon the backup registers all the time even reset.

The PCB design of STM32 MUC in the project used is shown in figure 3.7



3.3.2 Introduction of DHT 11 humidity & temperature sensor.

The DHT11 temperature and humidity sensor have a compound that calibrates the sensor for temperature and humidity output from the digital signal. High reliability and excellent long-term stability are provided using digital signal acquisition technology and exclusive temperature sensitivity technology. The sensor includes a resistance-type humidity measuring component and NTC temperature measuring components and is connected to a high-performance 8-bit microcontroller, providing excellent quality, fast response speed, relative capacity, and cost-effectiveness.

Each item is strictly DHT11 calibrated in the lab and is exactly accurate in moisture calibration. The calibration factor is stored as a program in OTP memory and is used by the internal sensor signal detection process. Single line serial interface is easily integrated quickly system. It is small, low in power consumption, and the signal transmission is 20 meters, which makes it a better choice for various applications, including the most demanding applications. This component is a 4-pin pack of pins. A proper connection and special packaging can be provided according to users' requests.

The acquisition of temperature and humidity data signals is achieved by the DHT11 module. This module is a digital temperature and humidity sensor with only one data cable that performs communication functions via a single bus protocol. The 5V power supply can be connected to the module circuit via the R1 resistor to improve the stability of this circuit. When the output of the sensor is low, the lower level is pulled down at the foot of the DATA tube inside the sensor, and the output height is pulled up to the high level, as shown in **Figure 3.8[b]** of the external circuit diagram of the DHT11 temperature and humidity sensor

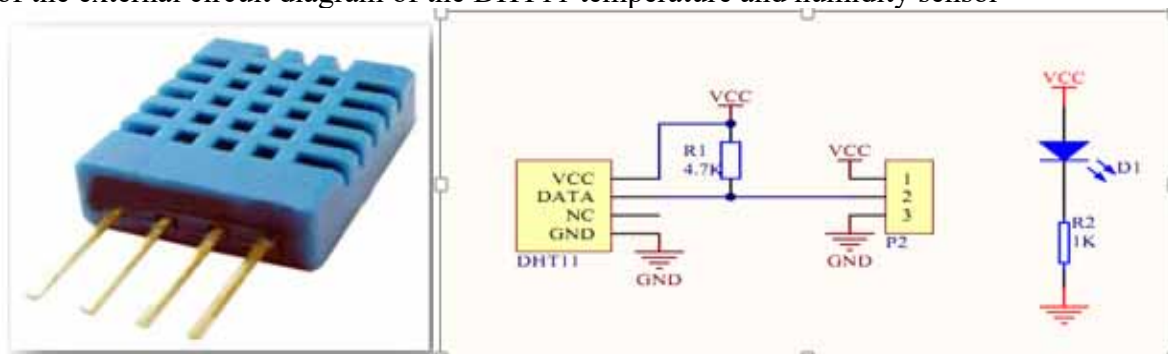


Figure 3.8 [a] DHT 11 Humidity & Temperature Sensor. Figure 3.8(b) DHT11 External circuit diagram of the temperature and humidity sensor

The DHT11 temperature and humidity sensor is a composite sensor that measures both temperature and humidity in the environment with two internal sensing devices, humidity sensor and temperature sensing device. The circuit parameters inside the sensor have been adjusted at the beginning of the design, the coefficients after calibration are stored in its internal storage space, and these data parameters can be called several times during subsequent development.

After successful sensor communication, a set of data is transmitted in the order of high and low. This set of data is 40 bits, respectively: 8 bits of humidity integer part, 8-bit humidity fraction part, 8-bit temperature integer part, 8-bit temperature fraction part, 8-bit check sum. You can observe the data measured by the temperature sensing device and the data measured by the humidity sensing device, compare the sum of the two with the data measured by the last

standard measurement device, see if it is equal, you can verify the correctness of the data read, or you can compensate for debugging according to the difference of the data.

3.3.3 Technical Specifications:

Table 3.4 overview

Item	Measurement Range	Humidity Accuracy	Temperature Accuracy	Resolution	Package
DHT11	20-90%RH 0-50 °C	±5%RH	±2°C	1	4 Pin Single Row

3.3.4 Detailed Specifications:

Table 3.5 Detailed Specifications

Parameters	Conditions	Minimum	Typical	Maximum
Humidity				
Resolution		1%RH	1%RH	1%RH
			8 Bit	
Repeatability			±1%RH	
Accuracy	25°C		±4%RH	
	0-50°C			±5%RH
Interchangeability	Fully Interchangeable			
Measurement Range	0°C	30%RH		90%RH
	25°C	20%RH		90%RH
	50°C	20%RH		80%RH
Response Time (Seconds)	1/e (63%)25°C, 1m/s Air	6 S	10 S	15 S
Hysteresis			±1%RH	
Long-Term Stability	Typical		±1%RH/year	
Temperature				
Resolution		1°C	1°C	1°C
		8 Bit	8 Bit	8 Bit
Repeatability			±1°C	
Accuracy		±1°C		±2°C
Measurement Range		0°C		50°C
Response Time (Seconds)	1/e (63%)	6 S		30 S

3.3.5 Typical Application.

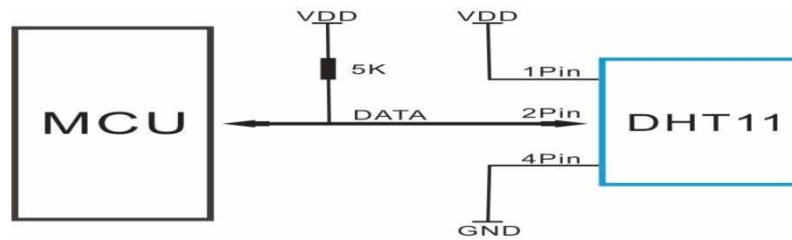


Figure 3-9 Typical Application

Note: 3Pin – Null; MCU = Micro-computer Unite or single chip Computer

When the connecting cable is shorter than 20 meters, a 5K pull-up resistor is recommended; when the connecting cable is longer than 20 meters, we choose an appropriate pull-up resistor as needed.

3.3.6 Power and Pin.

The DHT11 power supply is 3-5.5V DC. When power is supplied to the sensor, no instructions are transmitted to the sensor for one second, due to an unstable state. A capacitor, estimated at 100, can be added to the filter between VDD and GND.

3.3.7 Communication Process: Serial Interface (Single-Wire Two-Way)

The simple page data format is used to process communication and synchronization between the MCU and DHT11 sensors. A communication process is about 4mm. The data is made up of decimal and components. All data transfer is 40 bits, and the sensor sends a higher batch of data.

3.3.8 Overall Communication Process.

When the MCU sends an initial signal, from DHT11 Low Power Mode to operation mode, wait for the beginning of the MCU end signal. After it completes, the DHT11 is sent, which includes a 40-bit data response signal for the MCU relative humidity information and temperature. Users can choose some of the collected data (read). It will not give you the MCU reply signal of the MCU DHT11 signal. After collecting data, DHT11 will switch to low power mode until it does not receive an initial signal from the MCU.

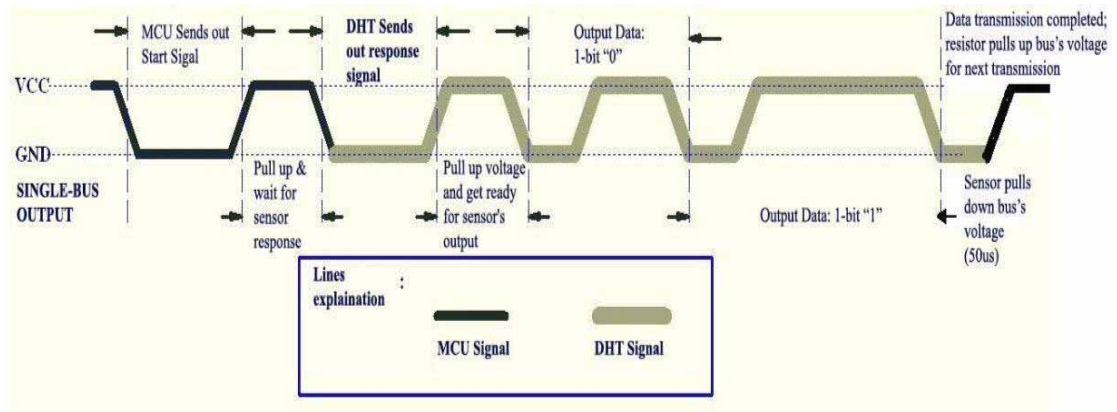


Figure 3 10 Overall Communication Process

3.3.9 MCU Sends out Start Signal to DHT

A data bus idle state is at a high voltage level. When the connection between the MCU and DHT11, the MCU sets the same voltage level data to a low level, and this process should take at least 18 milliseconds to ensure that the MCU DHT signal is detected, the MCU will tighten the voltage of wait 20 - 40 United States responded to DHT.

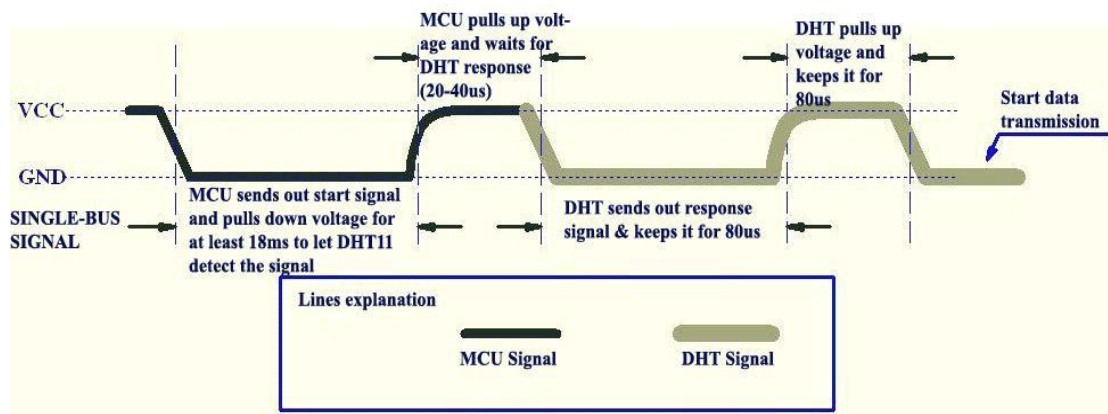


Figure 3 11 MCU Sends out Start Signal & DHT Responses.

3.3.10 DHT Responses to MCU

After DHT detects the initial signal, the response signal is transmitted at a low voltage level for 80US. Then the DHT program is set high with the bus voltage low and saves your 80Us to prepare the DHT data.

When the data on the same bus is at low voltage, this means that the DHT sends a response signal. After the DHT is sent to the answer signal, the voltage is extracted and saved as 80US and preparing for data transmission.

When the DHT sends data to the MCU, each data batch starts at low-pressure microseconds, and determines the length of the next high voltage signal that is the data When

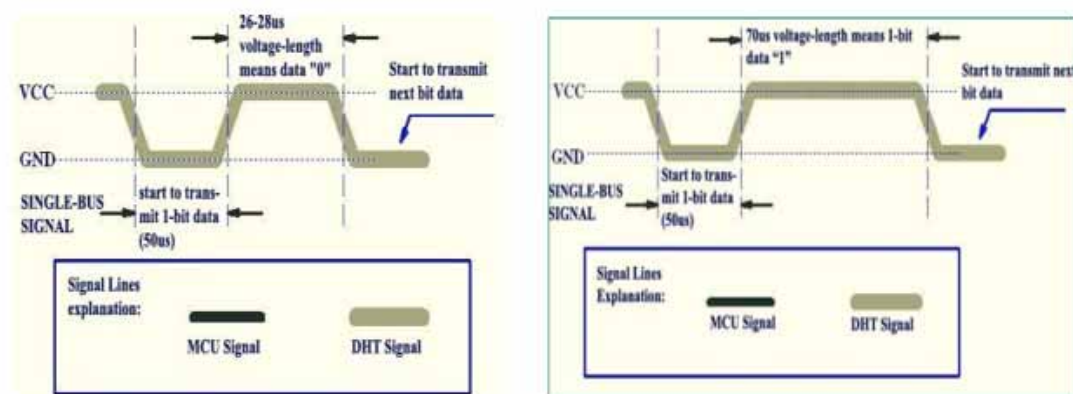


Figure 3.12 [a] Data "0" Indication Figure 3.12[b] Data "1" Indication

DHT sends data to the MCU, each bit of the data begins with a low-voltage microsecond, and the length of the next high-voltage level signal determines whether the data is "0" or "1".

If the DHT response signal is always at the high voltage level, the DHT response is said to be incorrect, and the connection is checked. When the last batch of data is transmitted, the DHT11 is lowered, and stored as 50 microseconds. Then the Odnobus voltage will be tightened to the resistor to adjust it to the Free State.

3.3.11 Electrical Characteristics

VDD=5V, T = 25°C (unless otherwise stated)

Table 3.6 Electrical Characteristics

	Conditions	Minimum	Typical	Maximum
Power Supply	DC	3V	5V	5.5V
Current Supply	Measuring	0.5mA		2.5mA
	Average	0.2mA		1mA
	Standby	100uA		150uA
Sampling period	Second	1		

Note: Sampling period at intervals should be no less than 1 second.

3.3.12 Attentions of application

1. Operating conditions

Use of the DHT11 sensor outside the operating range described in this data sheet may cause a 3 % RH to offset/match. When the DHT11 sensor returns to its normal working state and works within its limits, it can slowly return to calibration.

Please note that the use of the DHT11 sensor under unusual operating conditions will accelerate the aging process of the sensor.

2. Attention to chemical materials.

Chemical vapors can interfere with DHT sensing elements and degrade. Sensitivity. Prominent levels of chemical contamination can permanently damage the sensor.

3. Restoration process when (1) & (2) happen

Step 1: Place the DHT sensor at 50 ~ 60° C and humidity <10 %% RH for 2 hours. Step 2: Place the K DHT sensor at 20 ~ 30°C C temperature and humidity> 70 % RH for 5 hours.

4. Temperature Affect.

The relative humidity is highly dependent on temperature. Although temperature compensation technology is used to ensure accurate relative humidity measurements, it is highly recommended that the humidity and temperature sensors operate at the same temperature. DHT11 should be installed as much as possible from the heat-generating parts.

5. Light Affect

Prolonged exposure to sunlight and ultraviolet rays will reduce the effectiveness of DHT.

6. Connection wires

The quality of the contact wire will affect the quality and distance of the communication, so it is recommended to use a high-quality shielded wire.

7. Other attentions

- Soldering temperature should be less than 260 degrees Celsius, and contact time should be less than 10 seconds.
- Avoid using sensors in dew conditions.
- Do not use this product in a safety or emergency stop device, or in any other situation where DHT11 failure may cause personal injury.
- Storage: Store the sensor at 10-40 stores, relative humidity <60 stores.

3.4 IR Infrared Obstacle Avoidance Sensor Module.

Sensor module adapts to ambient light. There are a pair of infrared emitter tubes and receiver tubes. The emitter tube emits infrared radiation of a certain frequency when it detects the direction of the obstacle (reflective surface). Infrared radiation reflected by the receiver tube and processed by the comparator circuit is processed. From that point forward, the green light is on, yet the yield signal from the yield interface yields a computerized signal (low-level sign), the potentiometer of the location distance handle can be changed, the compelling distance range is 2 ~ 80cm, the functioning voltage is 3.3V-5V, and the sensor discovery reach can be changed The potentiometer is gotten, quick and simple to set up no manual intercession and easy to utilize. It may be utilized for robot snag evasion, vehicle hindrance aversion, line counting, highly contrasting line following, and more capacities on different events.

3.4.1 Working principle.

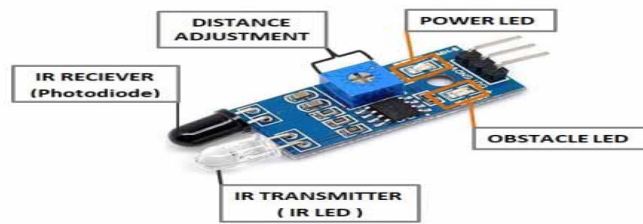


Figure 3 13 IR Infrared Obstacle Avoidance Sensor Module.

The infrared transmitter emits an infrared signal. In the case of a reflective surface (e.g., white color), the signal will be reflected in multiple directions, including the direction of the infrared receiver, and the infrared receiver receives the signal from the detected object. At the point when the surface is permeable (like dark), the infrared sign is not reflected, and the sensor cannot identify the article. This will be the case regardless of whether the thing is lost.

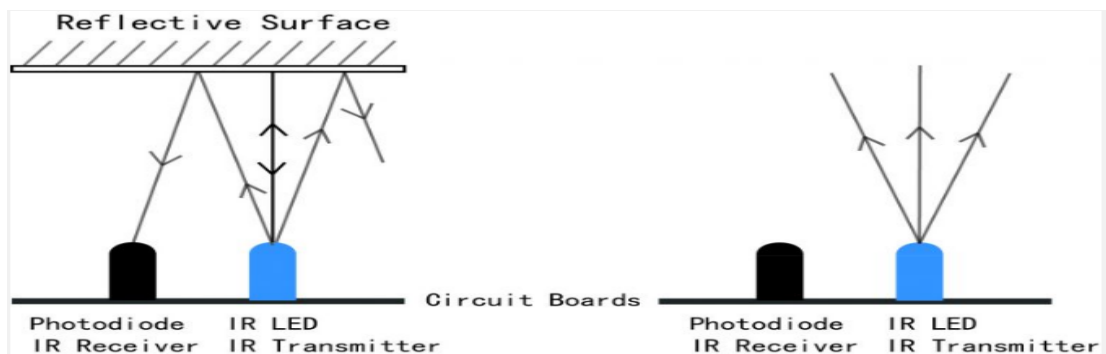


Figure 3 14 working mechanism IR Sensor Module.

3.4.2 Features of IR Sensor Module.

When the module detects an obstacle in front of the signal emitting by module, the green indicator light level on the PCB and the OUT port continues to output a low-level signal, the detection distance of the module is 2 ~ 80cm, the angle of detection is 35 °, the detection distance can be adjusted clockwise The potentiometer adjusts the potential to increase the detection distance; Adjust the potentiometer counterclockwise to reduce the sensing distance.

- Active infrared detection sensor for reflection, target reflectivity, and shape of the key detection range. The smallest black detection range, the largest white detection range; the distance from a small area to the object is small and a large area is far from the large area.
- The OUT port of the sensor module can connect directly to the I/O port of the single-chip microcomputer, or directly to the 5V relay, connection mode: VCC-VCC; Earth-Earth; output-IO.
- The comparator used by LM393 is stable.
- You can use a 3-5V DC power module. When the power is turned on, the red power LED turns on.
- With 3mm screw holes, easy to install.
- When leaving the factory, each module has an adjustable comparator threshold voltage potentiometer, please do not adjust the potentiometer under exceptional circumstances.

- This module has 3 pins-Vcc, GND, and output. We also have 2 LEDs, one for the power supply, which will light up when connected to the power supply. The other is for obstacle detection.
- Easy to assemble and use.
- Integrated detection instructions.
- Preset knobs to fine-tune the distance range.
- There is an obstacle, a green LED on the PCB.
- Adjustable detection range with the potentiometer.
- Clockwise: Increase the detection distance.
- Counterclockwise: decrease the detection distance.
- Circuit board size: 3.1 cm * 1.5 cm / 1.22 * 0.59 inches.

3.4.3 Connections.

- Connect the Vcc of the sensor module to the Vcc of the Arduino board.
- Connect the GND of the sensor module to the GND of the Arduino board.
- Connect the yield pin of the sensor module to stick 7 of the Arduino board.
- When the obstacle avoidance sensor detects an obstacle, the LED turns on. Otherwise, it will be disabled.
- With 3mm screw openings, simple to fix and introduce.

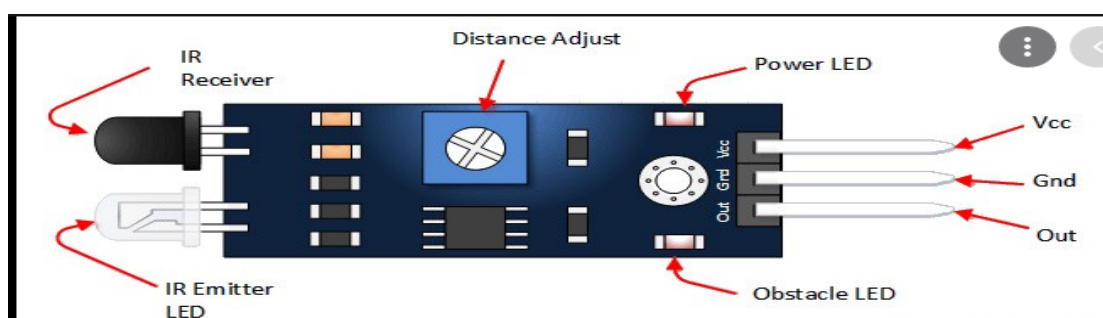


Figure 3 15 Connections IR Sensor Module.

3.4.4 Pin Description.

Pin, Control Indicator	Description
VCC	3.3 to 5 Vdc Supply Input
GND	Ground Input
OUT	Output that goes low when obstacle is in range
Power LED	Illuminates when power is applied
Obstacle LED	Illuminates when obstacle is detected
Distance Adjust	Adjust detection distance. CCW decreases distance. CW increases distance.
IR Emitter	Infrared emitter LED
IR Receiver	Infrared receiver that receives signal transmitted by Infrared emitter

3.4.5 Specification.

- When the module detects an obstacle to the signal, the green board-level indicator light is on, and the OUT port maintains a low output signal. The detection distance of the module is 2 ~ 80cm, the detection angle is 35 °, and the distance can be detected. The potentiometer can be adjusted clockwise to detect the distance. Increase, adjust the potentiometer counterclockwise to reduce the sensing distance.
- Active infrared detection sensor reflectivity, target reflectivity, so the shape is the key to the detection distance. The smallest detection range is black, white, and the largest; small items are away from small areas and large areas are away from large areas.
- The sensor module yield port OUT port can be straightforwardly associated with the microcontroller IO can likewise straightforwardly drive a 5V transfer; Connection: VCC-VCC; GND-GND; OUT-IO
- Comparators LM393, stable.
- The module can be a 3-5V DC power supply. At the point when the force is turned on, the red force pointer lights; with the screw openings 3mm, simple fixed establishment.
- Board size: 3.2CM * 1.4CM.
- Every module has been transported limit comparator voltage changed by potentiometer great, non-extraordinary case, do not flexible potentiometer.
- The yield port OUT sensor module can be straightforwardly associated with the microcontroller IO port; you can straightforwardly drive a 5V transfer.
- The module identifies the distance 2 ~ 30cm, discovery point 35 °, the distance can identify potential is changed.
- At the point when the force is turned on, the red force marker light.

3.4.6 Module Interface Description.

- VCC: 3.3V-5V external voltage (can be directly connected to 5v and 3.3v MCU)
- GND: GND External
- OUT: small board digital output interface (0 and 1)

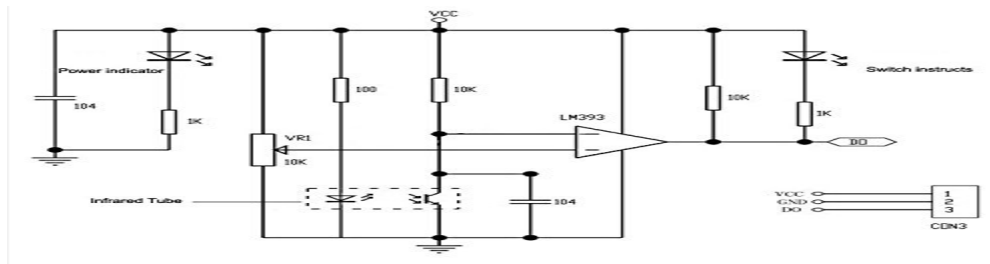


Figure 3 16 IR sensor Module Interface Description.

3.5 Flame / Fire sensor LM393 Module.

A fire sensor module that comprises of a fire sensor, resistor, capacitor, potentiometer, and comparator LM393 is an incorporated circuit. It can recognize infrared light with a frequency going from 700nm to 1000nm. The far-infrared fire test changes over the strength changes of the outside infrared light into current changes.

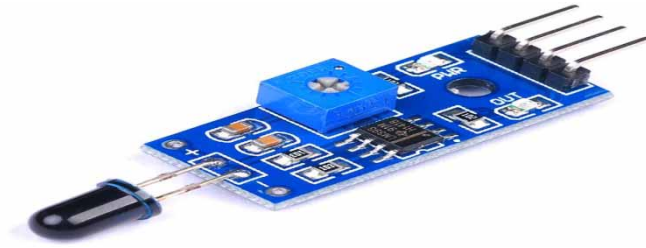


Figure 3 17 LM393 Module.

The LM193, LM293, and LM393 gadgets comprise of two free low voltage comparators planned explicitly to work from a solitary stockpile over a wide scope of voltages. Activity from split force supplies is likewise conceivable. These comparators likewise have a remarkable trademark in that the information normal mode voltage range incorporates ground even though worked from a solitary force supply voltage.

3.5.1 Working principle of a flame sensor.

Fire sensors are built with an electronic circuit that gets electromagnetic radiation. The sensor involves a segregated metal bar that coordinates a substituting current sign into the fire.

The fire gets this substituting current and infuses its immediate current part into the burner.

The burner is then answerable for leading the got dc signal first to the burner rack and afterward to the control by utilizing on solid land wire.

The control then, at that point, identifies the current circle and locks on the gas valve to proceed with the run. This profits as a little amperage/voltage signal that has high possibilities of sign debasement.

The fire finder follows the infrared fire streak technique that makes it equipped for working through a covering of residue, oil, water fume, and on occasion ice.

3.5.2 Types of Flame sensors.

1 Ultraviolet (UV) detectors.

They can identify flames and blasts in around 3-4 ms. The UV finder can precisely recognize the kind of fire. The drawback is that these indicators identify even daylight, radiation, bend welding, and lightning, and short-wave UV cannot enter dust, smoke, fuel particles, and oil films.

2 Infrared flame detectors.

These sensors work by utilizing an infrared band. A little warm imaging camera in these identifiers can in a split second distinguish the presence of hot gasses delivered anyplace close to the infrared indicator. In specific cases, different wellsprings of hot gas that happen close to the infrared fire identifier can make it set off a bogus alert. Infrared waves can enter dust, smoke, fuel particles, and oil films.

3 UV/IR detectors.

A mix of both UV and infrared innovation is very well known for the discovery of warmth as the identifier can gather data from both an UV point of view and an infrared viewpoint. This assists with limiting bogus alerts.

4 IR/IR flame detectors.

These recognize blazes inside two infrared frequencies.

3.5.3 Closed circuit video cameras.

They can recognize any fire or fire. Nevertheless, the opportunities for bogus cautions are extremely high as the cameras can be set off by any sort of smoke or mist. This sort of finder is not entirely dependable.

Aside from the UV, IR, apparent, and IR3 fire finders, a few different identifiers are likewise discovered; like 3IR+UV fire locators, IR/IR fire indicators, UV/IR fire finders, and Infrared warm cameras.

3.5.4 Detection range of a flame sensor.

The scope of discovery of a fire sensor relies upon where it is mounted, the openness of the indicator, and the encompassing environment of the finder. The scope of fire recognition is displayed in the picture beneath.

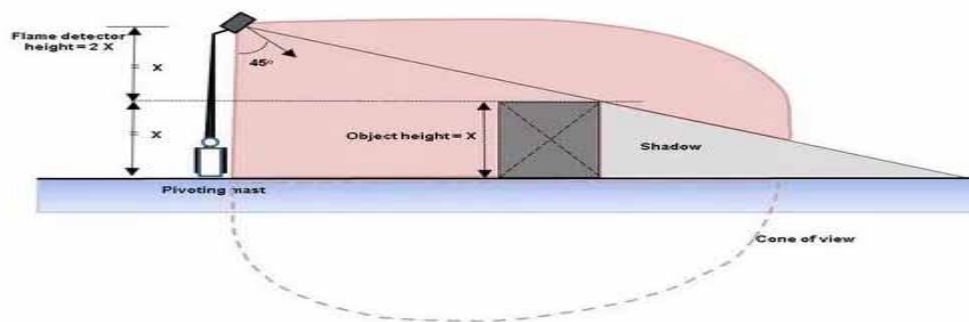


Figure 3 18 Detection range of a flame sensor

The fire identifier is set at a stature that is double the item. The downturn point is fixed at 45 degrees for most extreme reach inclusion,

3.5.5 Uses of fire sensors.

- Industrial gas turbines.
- Hydrogen stations.
- Fire recognition gadgets.
- Fire cautions.
- Domestic and mechanical warming frameworks.
- Firefighting robots.
- Industrial heaters.
- Drying frameworks.
- Gas-controlled cooking gadgets.
- Radiation research labs.
- Mines.
- Ignition frameworks.

- ### 3.5.6 Module Features and Specifications.

- The sensor and the fire ought to be kept at a specific distance to keep away from harm to the sensor at high temperatures. The fire distance of the lighter is 80cm. The bigger the fire, the farther the test distance is.
- Little board simple yield mode and AD change handling can accomplish higher accuracy.

3.5.9 Pin connections.

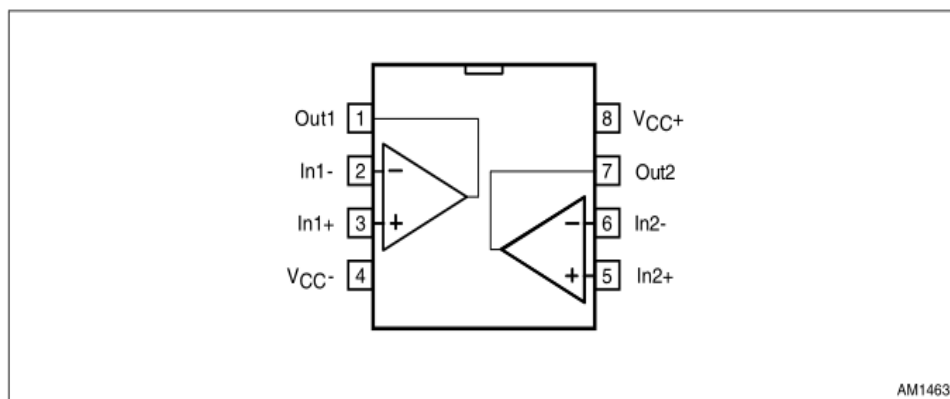


Figure 3 20 Pin connections (top view)

3.5.10 Electrical characteristics.

Table 3.7 $V_{CC+} = +5V$, $V_{CC-} = 0V$, $T_{amb} = +25\text{ }^{\circ}C$ (unless otherwise specified)

Symbol	Parameter	LM193A - LM293A LM393A			LM193- LM293 LM393			Unit
		Min.	Typ.	Max.	Min	Typ.	Max.	
V_{io}	Input offset voltage ⁽¹⁾ $T_{min} \leq T_{amb} \leq T_{max}$		1	2 4		1	5 9	mV
I_{io}	Input offset current $T_{min} \leq T_{amb} \leq T_{max}$		3	25 100		3	50 150	nA
I_{ib}	Input bias current (I^+ or I^-) ⁽²⁾ $T_{min} \leq T_{amb} \leq T_{max}$		20	100 300		20	250 400	nA
A_{vd}	Large signal voltage gain $V_{CC} = 15\text{ V}$, $R_L = 15\text{ k}\Omega$, $V_o = 1\text{ V}$ to 11 V	50	200		50	200		V/mV
I_{CC}	Supply current (all comparators) $V_{CC} = +5\text{ V}$, no load $V_{CC} = +30\text{ V}$, no load		0.45 0.6	1 2.5		0.45 0.6	1 2.5	mA
V_{id}	Differential input voltage ⁽³⁾			V_{CC}^+			V_{CC}^+	
V_{OL}	Low level output voltage $V_{id} = -1\text{ V}$, $I_{sink} = 4\text{ mA}$ $T_{min} \leq T_{amb} \leq T_{max}$		80	400 700		80	400 700	mV
I_{OH}	High level output current $V_{CC} = V_o = 30\text{ V}$, $V_{id} = 1\text{ V}$ $T_{min} \leq T_{amb} \leq T_{max}$		0.1	1		0.1	1	nA μA
I_{sink}	Output sink current $V_{id} = 1\text{ V}$, $V_o = 1.5\text{ V}$	6	18		6	18		mA
t_{re}	Response time ⁽⁴⁾ $R_L = 5.1\text{ k}\Omega$ connected to V_{CC}^+		1.3			1.3		μs
t_{rel}	Large signal response time $R_L = 5.1\text{ k}\Omega$ connected to V_{CC}^+ $e_1 = \text{TTL}$, $V_{(ref)} = +1.4\text{ V}$		300			300		ns

- At yield switch point, $V_o \approx 1.4\text{ V}$, $R_s = 0$ with V_{CC}^+ from 5 V to 30 V , and over the full normal mode range (0 V to $V_{CC}^+ - 1.5\text{ V}$).
- The bearing of the info current is out of the IC because of the PNP input stage. This current is steady, autonomous of the condition of the yield, so no stacking charge exists on the reference of information lines.
- Positive outings of info voltage might surpass the force supply level. However long the other voltage stays inside the normal mode range, the comparator will give an appropriate yield state. The low info voltage state should not be exactly -0.3 V (or 0.3 V beneath the negative force supply, whenever utilized).
- The reaction time determined is for a 100-mV input venture with a 5-mV overdrive. For bigger overdrive signals 300 ns can be gotten.

3.6. MQ2 Gas Sensor Module.



Figure 3 21 MQ2 Gas Sensor Module.

MQ2 gas sensing element may be wont to discover the presence of LPG, propane, and gas similarly as alkane series and different inflammable vapors, low value and appropriate for numerous applications. The sensing element is sensitive to inflammable gas and smoke. Offer five volts to power the smoke sensing element. The smoke sensing element displays smoke and voltage. Excessive smoke. A potentiometer is provided to regulate the sensitivity. SnO₂ may be a sensing element that has low physical phenomenon once the air is evident. However, once there's smoke, the sensing element can offer analog resistance output supported by the smoke density. There is a heater within the circuit. The heater is steam-powered by VCC and GND. The circuit has variable resistance. The resistance on the pin depends on the smoke within the air on the sensing element. If the fabric is high, the resistance can decrease. And the voltage between the sensing element and the load resistance will increase.

3.6.1 Working Principle.

MQ2 has an electrochemical sensor, which can change its resistance to different gases with different concentrations. The sensor is connected in series with the variable resistor to create a voltage divider circuit, and the variable resistor is used to change the sensitivity. When one of the above gas elements meets the sensor after heating, the resistance of the sensor changes. The change in resistance changes the voltage across the sensor, which the microcontroller can read. Knowing the reference voltage and the resistance value of another resistor, the voltage value can be used to determine the resistance value of the sensor. Sensors have different sensitivities to several types of gases.

Smoke sensing element in an alarm system is meant to discover suspicious smoke. Though ionization tube smoke alarms area units are offered, they are not allowed to be utilized in non-public homes because of they contain hot parts. Currently, the sole alarm system for home use is an associate optical alarm (also known as a physical phenomenon smoke alarm).

3.6.2 Why install a smoke alarm in your home?

Alarm system is a crucial issue to guard your home against the hearth. Since 2015, smoke alarms are obligatory, and each French home should have a minimum of one device that mechanically detects smoke and activates the alarm. The law was passed in March 2010 and applies to any or all homes: homes, apartments, volume housing, seasonal rental properties, etc. persona non grata The law needs smoke alarms to hold the cerium mark and befits European customary linear unit 14604.

That is why all owners ought to come upon smoke alarms. If the living accommodations are rented, each tenant and the landholder should confirm that the alarm system meets the wants which the arrival works properly. The tenant then handles the bags, checks its operation often, and replaces the battery.

The danger of a fireplace in your house is abundant above you think that. In France, the hearth department responds to a minimum of 280,000 calls annually, and fires cause close to 800 deaths. Since the law mandates the installation of smoke alarms in each home, variety the amount the quantity} of home fires has attenuated by four-dimensional {and the and therefore the and additionally the} number of victims has also attenuated. In countries wherever such legislation has been in situ for an extended time, the quantity of deaths thanks to hearth has been reduced by five hundredths.

Once a fireplace breaks out, it is typically necessary to reply terribly quickly. The hearth unfolds apace, and the deadly smoke slowly engulfed the air, inflicting suffocation. Therefore, it is necessary to discover smoke early so residents have lots of time to induce out and apprise emergency services. Smoke alarms will warn you throughout the day or night.

The amount of the alarm is additionally a necessary issue that affects its effectiveness. Most fatal fires occur at the hours of darkness once the relations area unit is asleep. Therefore, the audible alarm ought to be a minimum of eighty-five decibels and alert people that area unit asleep.

A standard idea is that smoke, smell, heat, or noise from a fireplace is enough to wake someone. Hearth produces extremely deadly fumes that cause a threat long before waking folks from the results of the fireplace (heat, noise). Smoke alarms can increase your possibilities of obtaining a timely alert and obtaining out of the building.

3.6.3 Basic Circuit.

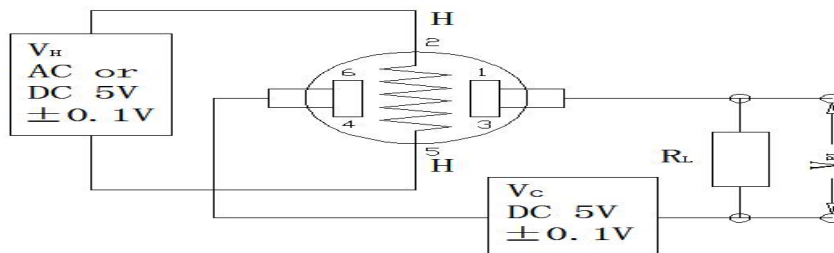


Figure 3 22 Basic Circuit of MQ2 Gas Sensor Module.

The on top of the image shows the fundamental look at the circuit of MQ-2. The sensing element needs 2 voltage inputs: heater voltage (VH) gate voltage (VC). The VH sensing element is employed to produce customary operating temperature and might settle for DC or AC power, whereas the load resistance nonparallel with the VRL sensing element is that the voltage of RL. VC load resistance provides associate elicited voltage for RL, and a DC power offer should be used.

3.6.4 Specifications.

- Power Supply: 4.5V to 5V DC
- High sensitivity to Propane, Smoke, LPG, and Butane
- Wide range high sensitivity to Combustible gases
- Long life and low cost
- Analog and Digital output available
- Onboard visual indicator (LED) for indicating alarm
- Compact design and easily mountable
- Simple 4 PIN header interface

- Drive circuit is simple.
- Sensor Type: Semiconductor
- Concentration: 300-10000ppm (Combustible gas)
- Supply voltage =5v

3.6.5 Applications.

- Safety of home
- Control of air quality
- Measurement of gas level
- Domestic gas leakage detector
- Industrial Combustible gas detector
- Portable gas detector

3.6.6 Structure and configuration.

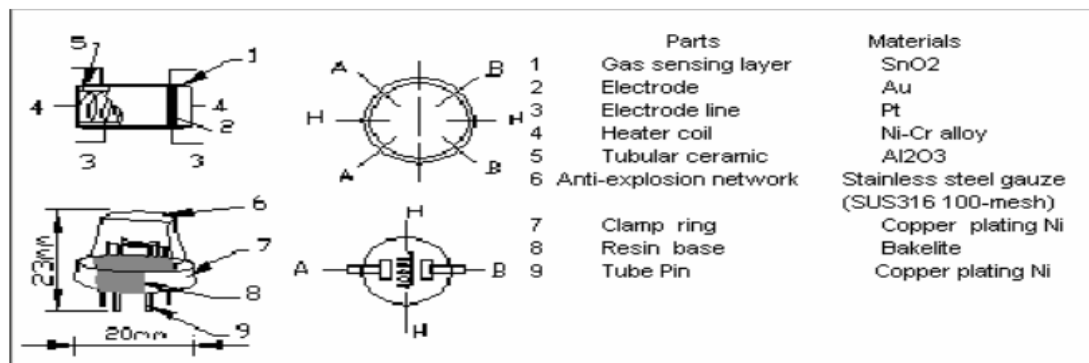


Figure 3 23 Structure and configuration of MQ2 Gas Sensor Module.

The sensor consists of a small AL₂O₃ ceramic tube, a ton of dioxide (SnO₂) sensing layer, a measuring electrode, and a heater, and is fixed in a plastic shell and stainless-steel mesh. The heater provides the necessary operating conditions for sensitive components. The shell MQ-2 has 6 pins, 4 of which are used for receiving signals, and the other 2 are used for heating current.

3.6.7 Basic test loop.

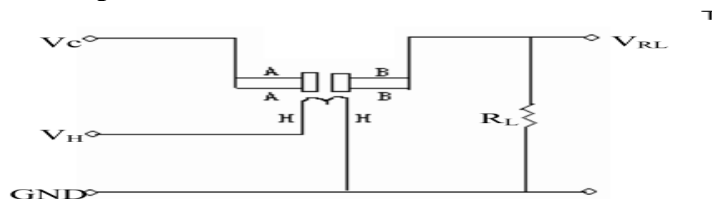


Figure 3 24 Basic test loop of MQ2 Gas Sensor Module.

The above is basic test circuit of the sensor. The sensor needs to be put 2 voltage, heater voltage (VH) and test voltage (VC). VH used to supply certified working temperature to the sensor, while VC used to detect voltage (VRL) on load resistance (RL) whom is in series with sensor. The sensor has light polarity, and Vc need DC power. VC and VH could use same power circuit with precondition to assure performance of sensor. To make the sensor with better performance, suitable RL value is needed:

Power of Sensitivity body (Ps):

$$Ps = Vc2 \times \frac{Rs}{(Rs + RL)2}$$

$$Ressistace\ of\ sensor\ (Rs): Rs = \left(\frac{Vc}{VRL} - 1 \right) * RL$$

Table 3.8 Technical Data.

Model No			MQ-2
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite (Black Bakelite)
Detection Gas			Combustible gas and smoke
Concentration			300-10000ppm (Combustible gas)
Circuit	Loop voltage	Vc	≤24V DC
	Heater Voltage	VH	5.0V±0.2V AC or DC
	Load Resistance	RL	Adjustable
Character	Heater Resistance	RH	31Ω±3Ω (Room Tem.)
	Heater Consumption	PH	≤900mW
	Sensing Resistance	Rs	2KΩ-20KΩ (in 2000ppm C3H8)
	Sensitivity	S	Rs (in air)/Rs (1000ppm isobutane) ≥5
	Slope	α	≤0.6(R5000ppm/R3000ppm CH4)
Condition	Tem. Humidity		20°C±2°C; 65%±5%RH
	Standard test circuit		Vc:5.0V±0.1V; VH: 5.0V±0.1V
	Preheat time		Over 48 hours

3.6.8 Sensitivity Characteristics.

- Good sensitivity to Combustible gas in wide range
- High sensitivity to LPG, Propane and Hydrogen
- Long life and low cost
- Simple drive circuit

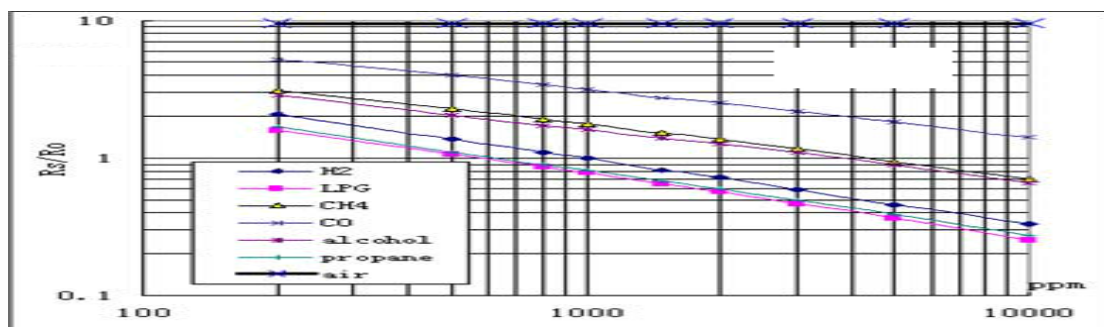


Figure 3 25 typical sensitivity characteristics of MQ2 Gas Sensor Module.

The ordinate MQ-2 represents the ratio of sensor resistance (R / R_o) and the concentration of abscissa gas. Rupee represents resistance to various gases, and Rupee represents the resistance of the 1000 ppm hydrogen sensor. All tests are performed under standard test conditions.

3.6.9 Influence of Temperature/Humidity.

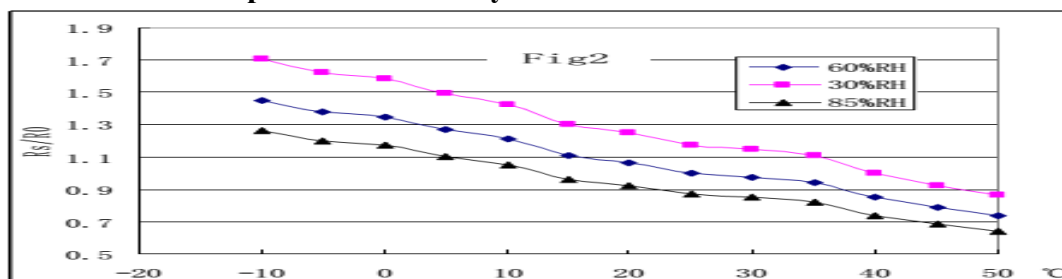


Figure 3 26 typical temperature and humidity characteristics.

The ordinate represents the sensor's resistance coefficient (R_O / R_O), and the RS represents the sensor's resistance in 1000ppm butane under another instruction. And moisture. R_o represents the resistance of the sensor in the medium. 1000 ppm methane, 20 °C / 65 R RH.

3.7 Wi-Fi module Introduction.

ESP-01 Wi-Fi module is developed by Ai-thinker Team. core processor ESP8266 in smaller sizes of the module encapsulates Ten silica L106 integrates industry-leading ultra-low power 32-bit MCU micro, with the 16-bit short mode, Clock speed support 80 MHz, 160 MHz, supports the RTOS, integrated Wi-Fi MAC/BB/RF/PA/LLNA, on-board antenna.

The module supports standard IEEE802.11 b/g/n agreement, complete TCP/IP protocol stack. Users can use the add modules to an existing device networking or building a separate network controller.

ESP8266 is high integration wireless SOCs, designed for space and power constrained mobile platform designers. it provides unsurpassed ability to embed Wi-Fi capabilities within other systems, or to function as a standalone application, with the lowest cost, and minimal space requirement.

ESP8266EX offers a complete and self-contained Wi-Fi networking solution; it can be used to host the application or to offload Wi-Fi networking functions from another application processor.

When ESP8266EX hosts the application, it boots up directly from an external flash. In has integrated cache to improve the performance of the system inn such applications.

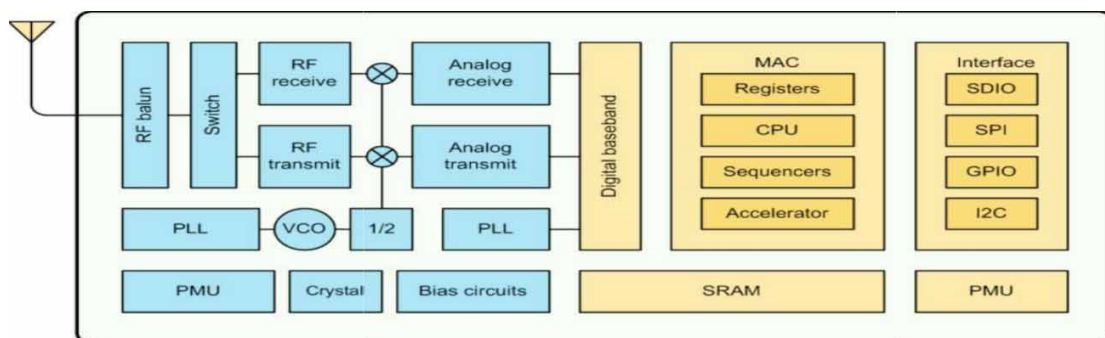


Figure 3 27 ESP8266EX Block Diagram

Alternately, serving as a Wi-Fi adapter, wireless internet access can be added to any micro controller-based design with simple connectivity (SPI/SDIO or I2C//UART interface

ESP8266EX is among the most integrated Wi-Fi chip in the industry; it integrates the antenna switches, RF balun, (balanced line to unbalanced line) power amplifier, low noise receive amplifier, filters, power management modules, it requires **minimal** external circuitry, and the entire solution, including front-end module, is designed to occupy minimal PCB area.

ESP8266EX also integrates an enhanced version of ten silica's L106 Diamond series 32-bit processor, with on-chip SRAM, besides the Wi-Fi functionalities. ESP8266EX is often integrated with external sensors and other application specific devices through its GPIOs; codes for such applications are provided in examples in the SDK.

Espressif Systems' Smart Connectivity Platform (ESCP) demonstrates sophisticated system-level features include fast sleep/wake context switching for energy-efficient VoIP, adaptive radio biasing. For low-power operation, advance signal processing, and spur cancellation and radio co-existence features for common cellular, Bluetooth, DDR, LVDS, LCD interference mitigation.

3.7.1 Packaging and Dimension.

The external size of the module is 14.3mm*24.8mm*3mm, as is illustrated in Figure 3.28[a] and figure 3.28[b] below. The type of flash integrated in this module is an SPI flash, the capacity of which is 1 MB, and the package size of which is SOP-210mil. The antenna applied on this module is a 3DBi PCB-on-board antenna.

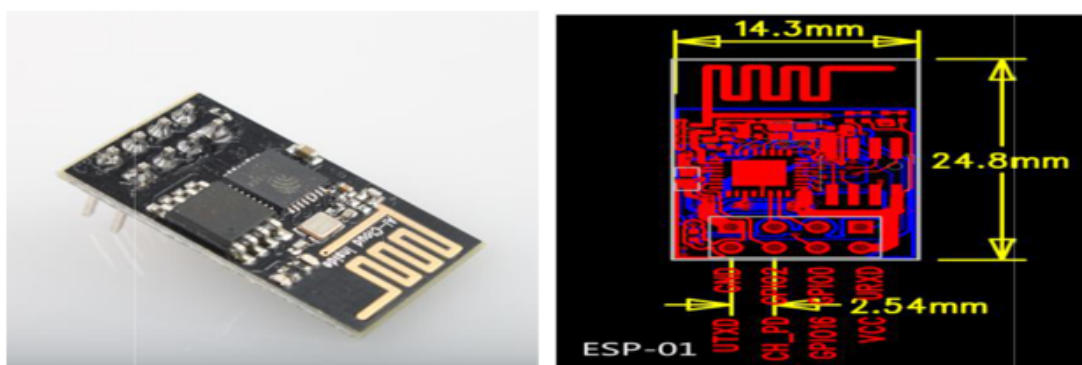


Figure 3 28 [a] [Module Pin Counts, 8 pin, 14.3 mm *24.8 mm *3.0 mm. Figure 3.28[b] Top View of ESP-01 Wi-Fi Module.

3.7.2 Schematics Diagram.

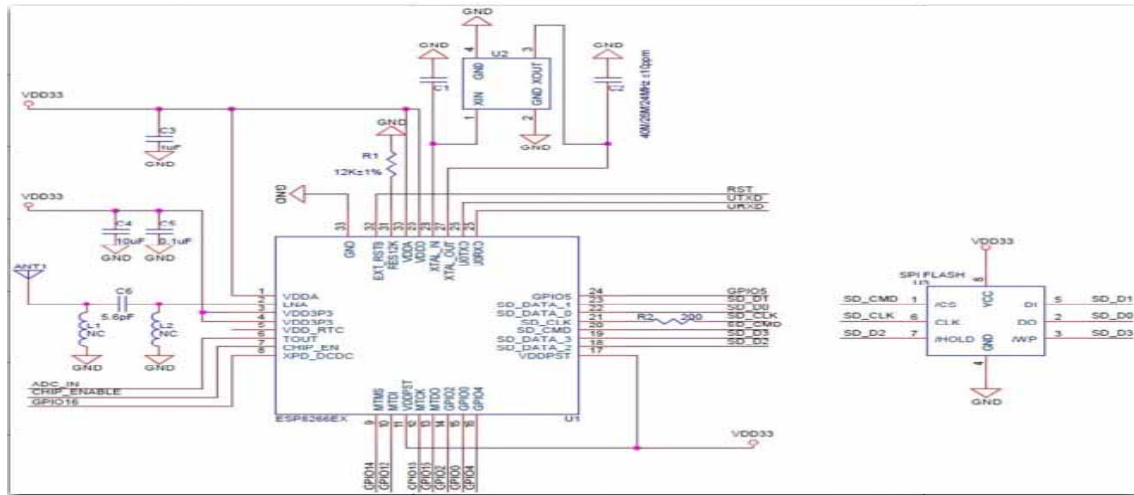


Figure 3 29 Schematics of Esp-01 Wi-Fi Module

There are three ways this wireless module works: the client mode, which transmits data information through the module to other wireless networks, the wireless access mode, which connects the module to other networks as a network site to transmit data to other devices, and the wireless-client mode, which can act as both the sender of the data and the network to receive data. In the design is the wireless and client mode, that is, the AP-STA mode, which can be used as a client or server can also be used as a commonly used Bluetooth connection module, in the configuration of IP address and port number, and other devices successfully docking, the two sides can transfer data to each other.

3.7.3 Features.

- 802.11 b/g/n.
- Integrated low power 32-bit MCU.
- Integrated 10-bit ADC.
- Integrated TCP/IP protocol stack.
- Integrated TR switch, balun, LNA, power amplifier and matching network.
- Integrated PLL, regulators, and power management units.
- Supports antenna diversity.
- Wi-Fi 2.4 GHz, support WPA/WPA2.
- Support STA/AP/STA+AP operation modes.
- Support Smart Link Function for both Android and iOS devices.
- Support Smart Link Function for both Android and iOS devices.
- SDIO 2.0, (H) SPI, UART, I2C, I2S, IRDAA, PWM, GPIO.
- STBC, 1x1 MIMO, 2x1 MIMO.
- A-MPDU & A-MSDU aggregation and 0.4s guard interval.
- Deep sleep power <10uA, Power down leakage current < 5uA.
- Wake up and transmit packets in < 2ms.
- Standby power consumption of < 1.0mW (DTIM3).
- +20dBm output power in 802.11b mode.
- Operating temperature range -40C ~ 12.

3.7.4 Parameters.

Table 3.10 describes the major parameters.

Categories	Items	Values
Wi-Fi Parameters	Wi-Fi Protocodes	802.11b/g/n
	Frequency Range	2.4GHz-2.5GHz(2400M-2483.5M)
Hardware Parameters	Peripheral Bus	UART/HSPI/12C/12S/Ir Remote control
		GPIO/PWM
	Operating Voltage	3.0~3.6V
	Operating Current	Average Value: 80mA
	Operating Temperature Range	-40 [®] ~125 [®]
	Ambient Temperature Range	Normal Temperature
	Package Size	14.3mm*24.8*3mm
	External Internal interface	N/A
Software parameters	Wi-Fi	Station/softAP/SoftAP+station
	Security	WPA/WPA2
	Encryption	WEP/TKIP/AES
	Firmware Upgrade	UART Download/OTA/ (via network)/download and write firmware via host
	Software Development	Support Cloud Server development /SDK for custom firmware development
	Network Protocols	IPv4, TCP/HTTP/FTP
	User Configuration	AT Instruction set, Cloud Server, Android /iOS App

3.7.5 Pin Descriptions.

Table 3.11 Pin Descriptions

NO.	Pin Name	Function
1	GND	GND
2	GPIO2	GPIO, Internal Pull-up
3	GPIO0	GPIO, Internal Pull-up
4	RXD	UART0, data received pin RXD
5	VCC	3.3V power supply (VDD)
6	RST	1) External reset pin, active low 2) Can loft or external MCU
7	CH_PD	Chip enable pin. Active high
8	TXD	UART0, data send pin RXD

3.7.6 Pin Mode.

Table 3.12 Pin Mode

Mode	GPPI015	GPIO0	GPIO2
UART	Low	Low	High
Flash Boot	Low	High	High

3.7.7 Receiver Sensitivity.

Table 3.13 Receiver Sensitivity.

Parameters	Min	Typical	Max	Unit
Input frequency	2412		2484	MHz
Input impedance		50		Ω
Input reflection			-10	dB
Output power of PA for 72.2Mbps	15.5	16.5	17.5	dBm
Output power of PA for 11b mode	19.5	20.5	21.5	dBm
Sensitivity				
DSSS, 1Mbps		-98		dBm
CCK, 11Mbps		-91		dBm
6Mbps (1/2 BPSK)		-93		dBm
54Mbps (3/4 64-QAM)		-75		dBm
HT20, MCS7 (65Mbps, 72.2Mbps)		-72		dBm
Adjacent Channel Rejection				
OFDM, 6Mbps		37		dB
OFDM, 54Mbps		21		dB
HT20, MCS0		37		dB
HT20, MCS7		20		dB

3.7.8 Functional Descriptions

▪ MCU.

ESP8266EX is embedded with ten silica L1106 32-bit micro controller (MCU), which features extra low power consumption and 16-bit RSIC. The CPU clock speed is 80MHz. It can also reach a maximum value of 160MHz. ESP8266EX is often integrated with external sensors and other specific devices through its GPIOs; codes for such applications are provided in examples in the SDK.

3.7.9 Memory Organization.

1. Internal SRAM and ROM.

ESP8266EX Wi-Fi SoC is embedded with memory controller, including SRAM and ROM. MCU can visit the memory units through I bus, d Bus, and AHB interfaces. All memory units can be visited upon request, while a memory arbiter will decide the running sequence according to the time when these requests are received by the processor.

According to our current version of SDK provided, SRAM space that is available to users is assigned as below:

- RAM size < 36kB when ESP8266EX is working under the station mode and is connected to the Router, programmable space accessible to user in heap and data section is around 36kB.)
- there is no programmable ROM in the SoC, therefore, user program must be stored in an external SPI flash.

2. External SPI Flash.

This module is mounted with a 1 MB external SPI flash to store user programs. If larger definable storage space is required, a SPI flash with larger memory size is preferred. Theoretically speaking, up to 16 MB memory capacity can be supported.

Suggested SPI Flash memory capacity:

- OTA is disabled: the minimum flash memory that can be supported is 512 kb.
- OTA is enabled: the minimum flash memory that can be supported is 1 MB.

Several SPI modes can be supported, including Standard SPI, Dual SPI, and Quad SPI.

Therefore, please choose the correct SPI mode when you are downloading into the flash, otherwise firmware's/programs that you downloaded may not work in the right way.

Crystal

Currently, the frequency of crystal oscillators supported include 40MHz, 26MHz and 24MHHz. The accuracy of crystal oscillators applied should be $\pm 10\text{PPPM}$, and the operating temperature range should be between -20°C and 85°C .

When using the downloading tools, please remember to select the right crystal oscillator type. In circuit design, capacitors C1 and C2, which are connected to the earth, are added to the input and output terminals of the crystal oscillator, respectively. The values of the two capacitors can be flexible, ranging from 6pF to 22pF, however, the specific capacitive values of C1 and C2 depend on further testing and adjustment on the overall performance of the whole circuit.

Normally, the capacitive values of C1 and C2 are within 10pF if the crystal oscillator frequency is 26MHz, while the values of C1 and C2 are $10\text{pF} < \text{C1}, \text{C2} < 22\text{pF}$ if the crystal oscillator frequency is 40MHz.

3.7.10 Interfaces.

Table 3.14 Descriptions of Interfaces.

Interface	Pin Name	Discription
HSPI	IO12(MISO) IO13(MOSI) IO14(CLK) IO15(CS)	SPI Flash 2,display screen,MCU can be connected using HSPI interface.
PWM	IO12(R) IO13(G) IO14(B)	Currently the PWM interface has four channelsbut users can extend the channels according to their own needs.PWM interface can be used to contro LED lights,buzzersrelayselectronic machines and so on.
IR Remote Control	IO14(R_T) IO5(IR_R)	The functionality of Infrared remote control interface can be implemented via software programming.NECcodingmodulationand demodulation are used by this interface.The frequency of modulated carrier signal is 38KHz.
ADC	TOUT	ESP8266EXintegrates a 10-bitanalog ADC.It can be used to test the power supply voltage ofVDD3P3(Pin3and Pin4)and the input power voltage of TOUT(Pin 6).However,these two functions cannot be used simultaneously This interface is typically used in sensor products.
12C	IO14(SCL) IO2(SDA)	12C interface can be used to connect external sensor products and display screensetc.
UART	UART0: TXD (U0TXD) RXD (U0TXD) IO15(RTS) IO13(CTS) UART1: IO2(TXD)	Devices with UART interfaces can be connected with the module Downloading:U0TXD+U0RXDorGPIO2+URXD Communicating:UART0:U0TXD.UORXD.MTDO(UORTS)MTCK(UOCTS Debugging:UART1_TXD(GPIO2)can be used to print debugging information.
		By defaultUART0 will output some printed information when the device is powered on and is booting up. If this issue exerts influence on some specific applicationsusers can exchange the inner pins of UARTwhen initializing that is to say,exchange U0TXD,URXD with URTSUCTS
12S	12S Input: IO12(12SI_DATA); IO13(12SI_BCK); IO14(12SI_WS); 12S Output: IO15(12SO_DATA); IO13(12SO_BCK); IO14(12SO_WS);	12S interface is mainly used for collecting ,processing ,and transformation of audio data

3.7.11 Absolute Maximum Ratings.

Table 3.15 Absolute Maximum Ratings

Rating	Condition	Value	Unit
Storage Temperature		-40 to 125	°C
Maximum Soldering Temperature		260	°C
Supply Voltage	IPC/JEDEC J-STD-020	+3.0 to +3.6	V

3.7.12 Recommended Operating Conditions.

Table 3.16 Recommended Operating Condition

Operating Condition	Symbol	Min	Typ	Max	Unit
Operating Temperature		-40	20	125	°C
Supply voltage	VDD	3.0	3.3	3.6	V

3.7.13 Digital Terminal Characteristics.

Table 3.17 Digital Terminal Characteristics

Terminals	Symbol	Min	Typ	Max	Unit
Input logic level low	V _{IL}	-0.3		0.25VDD	V
Input logic level high	V _{IH}	0.75VDD		VDD+0.3	V
Output logic level low	V _{OL}	N		0.1VDD	V
Output logic level high	V _{OH}	0.8VDD		N	V

Note: Test conditions: VDD = 3.3V, Temperature = 20 °C, if nothing special is stated.

3.7.14 RF Performance.

Table 3.18 RF Performance

Description	Min.	Typ.	Max	Unit
Input frequency	2400		2483.5	MHz
Input impedance		50		ohm
Input reflection			-10	dB
Output power of PA for 72.2Mbps	15.5	16.5	17.5	dBm
Output power of PA for 11b mode	19.5	20.5	21.5	dBm
Sensitivity				
CCK, 1Mbps		-98		dBm
CCK, 11Mbps		-91		dBm
6Mbps (1/2 BPSK)		-93		dBm
54Mbps (3/4 64-QAM)		-75		dBm
HT20, MCS7 (65Mbps, 72.2Mbps)		-72		dBm
Adjacent Channel Rejection				
OFDM, 6Mbps		37		dB
OFDM, 54Mbps		21		dB
HT20, MCS0		37		dB
HT20, MCS7		20		dB

3.7.15 Power Consumption.

Table 3.19 Power Consumption.

Parameters	Min	Typical	Max	Unit
Tx802.11b, CCK 11Mbps, P OUT=+17dBm		170		mA
Tx 802.11g, OFDM 54Mbps, P OUT =+15dBm		140		mA
Tx 802.11n, MCS7, P OUT =+13dBm		120		mA
Rx 802.11b, 1024 bytes packet length , -80dBm		50		mA
Rx 802.11g, 1024 bytes packet length, -70dBm		56		mA
Rx 802.11n, 1024 bytes packet length, -65dBm		56		mA
Modem-Sleep①		15		mA
Light-Sleep②		0.9		mA
Deep-Sleep③		10		uA

- Modem-Sleep requires the CPU to be working, as in PWM or I2S applications. According to 802.11 standards (like U-APSD), it saves power to shut down the Wi-Fi Modem circuit while maintaining a Wi-Fi connection with no data transmission. E.g., in DTIM3, to maintain a sleep 300mswake 3ms cycle to receive AP's Beacon packages, the current is about 15mA.
- During Light-Sleep, the CPU may be suspended in applications like Wi-Fi switch. Without data transmission, the Wi-Fi Modem circuit can be turned off and CPU suspended to save power according to the 802.11 standard (U-APSD). E.g., in DTIM3, to maintain a sleep 300ms-wake 3ms cycle to receive AP's Beacon packages, the current is about 0.9mA.
- Deep-Sleep does not require Wi-Fi connection to be maintained. For application with long time lags between data transmission, e.g., a temperature sensor that checks the temperature every 100s, sleep 300s and waking up to connect to the AP (taking about 0.3~1s), the overall average current is less than 1mA.

3.7.16 Reflow Profile.

Table 3.20 Instructions

T _S max to T _L (Ramp-up Rate)	3°C/second max
Preheat	
Temperature Min.(T _S Min.)	150°C
Temperature Typical.(T _S Typ.)	175°C
Temperature Min.(T _S Max.)	200°C
Time(T _S)	60~180 seconds
Ramp-up rate (T _L to T _P)	3°C/second max
Time Maintained Above: --Temperature(T _L)/Time(T _L)	217°C/60~150 seconds
Peak Temperature(T _P)	260°C max. for 10 seconds
Target Peak Temperature (T _P Target)	260°C +0/-5°C
Time within 5°C of actual peak(t _p)	20~40 seconds
T _S max to T _L (Ramp-down Rate)	6°C/second max
Tune 25°C to Peak Temperature (t)	8 minutes max

3.8 HYDZ Buzzer Module.

The ringer is a voice gadget that changes over a sound model into a signal. it is primarily used to provoke or caution. As per various plans and applications, it can create music sound, ringer, caution sound, electric chime, and other various sounds.

Regular applications incorporate an alarm, caution gadget, alarm, air protection alert, criminal caution, and clock, and so on it is broadly utilized in-home devices, caution frameworks, programmed creation lines, low voltage electrical gear, electronic toys, game machines, and different items and ventures.

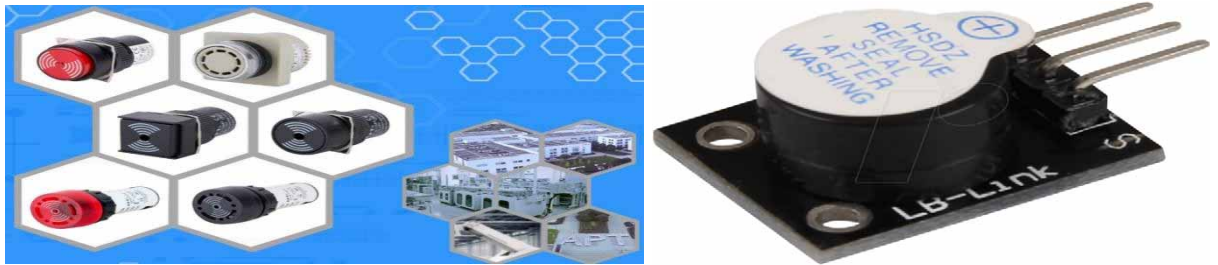


Figure 3 30 HYDZ Buzzers.

An HYDZ bell is a little yet productive part to add sound elements to our venture/framework. It is a tiny and reduced 2-pin structure that consequently can be effectively utilized on the breadboard, Perf Board, and surprisingly on PCBs which makes this a utilized part in most electronic applications.

There are two sorts of signals as per their sound activity that is usually accessible. The one displayed here is a basic bell which when fueled will make a Continuous Beweeps.... sound, the other kind is known as a readymade bell which will look bulkier than this and will create a Beep. Signal. Signal. Sound because of the inward swaying circuit present inside it. In any case, the one displayed here is most broadly utilized because it very well may be redne with the assistance of different circuits to fit effectively in our application.

This ringer can be utilized by driving it utilizing a DC power supply going from 4V to 9V. A straightforward 9V battery can likewise be utilized, yet it is prescribed to utilize a directed +5V or +6V DC supply. The signal is regularly connected with a changing circuit to wind down ON or turn the ringer at the necessary time and require stretch.

Buzzers vary in development, size, and detail. Doorbells of several types and sizes are used in different applications. Depending on the development, it is accompanied by several bells:

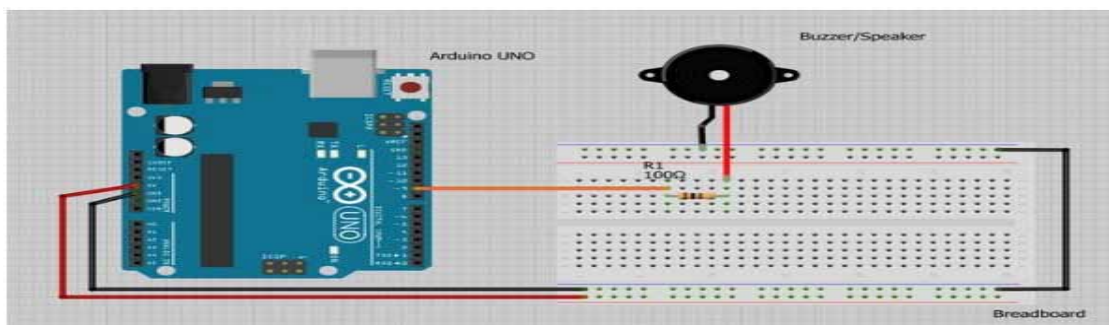


Figure 3 31 HYDZ Buzzer circuit connection Module.

i. Electromechanical.

The electromechanical doorbell was developed by Joseph Henry in 1831. It was used primarily in the first doorbells until they were discontinued in the mid-1930s due to their flowing tone.

The first devices were based on an electromechanical frame, which was no different from an electric bell without a metal gong. In the same way, switching can be associated with the penetration of the control current into itself, causing the contacts to vibrate (while the alternating current is regulated, the contacts will vibrate when the line repeats), as a rule, these blocks are placed in a divider. Or on the roof. Use as a discovery table. The "bell" comes from the vibration of the grille, emitted by the electromechanical buzzer.

ii. Mechanical bell.

Doorbells are simple mechanical signal icons that require conductors. Another example is a doorbell.

iii. Piezoelectric doorbell.

In some cases, piezoelectric bells or piezoelectric signals were designed by Japanese manufacturers and installed in a wide variety of exhibits from the 1970s to the 1980s. This progress is due to the active efforts of the Japanese assembly organization. In 1951, they formed the Barium Titanate Research Committee, which allowed these organizations to "seriously help" and made some breakthroughs and innovations in piezoelectric.

The piezoelectric element can be driven by an oscillating electronic circuit or other sound source and driven by a piezoelectric sound amplifier. The sound commonly used to indicate the pressing of a button is a tick, tap, or beep.

The piezoelectric buzzer/buzzer also relies on cavity reverberation or Helmholtz reverberation to provide a distinguishable buzzer. The piezoelectric buzzer/buzzer also relies on cavity reverberation or Helmholtz reverberation to provide a distinguishable buzzer.

3.8.1 Applications of Buzzer.

- Alarming Circuits, where the client must be frightened with regards to something
- Communication hardware
- Automobile gadgets
- Portable hardware, because of its minimized size
- Novelty employments
- Judging boards
- Educational purposes
- Annunciator boards
- Electronic metronomes
- Game show lock-out gadget
- Microwave stoves and other domestic devices
- sporting occasions, for example, b-ball games
- Electrical alerts
- Computer and peripherals
- Electronic sales register

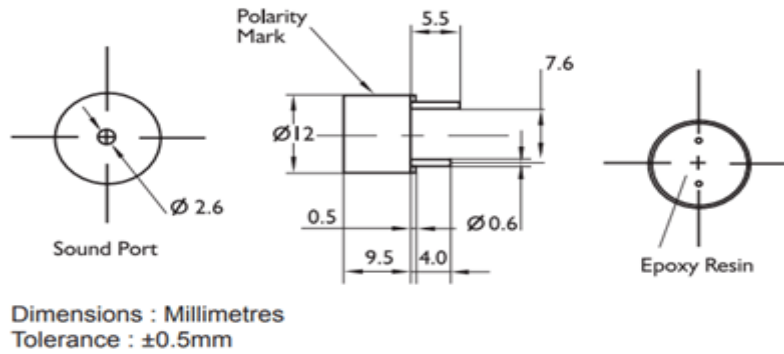


Figure 3.32 internal structure of buzzer module.

3.8.2 Buzzer Pin Configuration.

Table 3.21 Pin Configuration

Pin No	Pin Name	Description
1	Positive	Identified by (+) symbol or longer terminal lead. Can be powered by 6V DC
2	Negative	Identified by short terminal lead. Typically connected to the ground of the circuit

3.8.3 Buzzer Features and Specifications.

- Rated Voltage: 6V DC
- Operating Voltage: 4-8V DC
- Rated current: $<30\text{mA}$
- Sound Type: Continuous Beep (at 10cm^* : $\geq 85\text{d}$)
- Resonant Frequency: $\sim 2300\text{ Hz}$
- Small and perfect fixed bundle
- Breadboard and Perf board well disposed
- Black
- With inward drive circuit
- Sealed structure
- Wave solder able and launder able
- Operating Temperature: -25°C to $+80^\circ\text{C}$
- Storage Temperature: -30°C to $+85^\circ\text{C}$
- Weight: 2g

3.9 128*64 OLED.

OLED is an abbreviation Organic light emitting diodes are a kind of light emitting diodes (LEDs) in which the light emitting layer is an organic compound film that emits light in response to current. The organic semiconductor layer is usually installed in the middle of two electrodes more than one of which is transparent. The OLED I used in my project design is 128*31 OLED to display the information of my project design. These displays are small, and its diagonal is about an inch but very readable due to the high contrast of OLED displays. The display consists of 128x32 separate white OLED pixels each of which is turned on or off by the controller chip because the display makes its own lights there is no need for backlighting.

This reduces the power required to run OLEDs which is why these displays have such high contrast. OLED pins can be chosen which will have a microcontroller interface including

serial peripheral interface. SSD1306 provides a platform on which various tasks can be implemented, including contrast control, display RAM and oscillator, reducing the number and power consumption of external components.

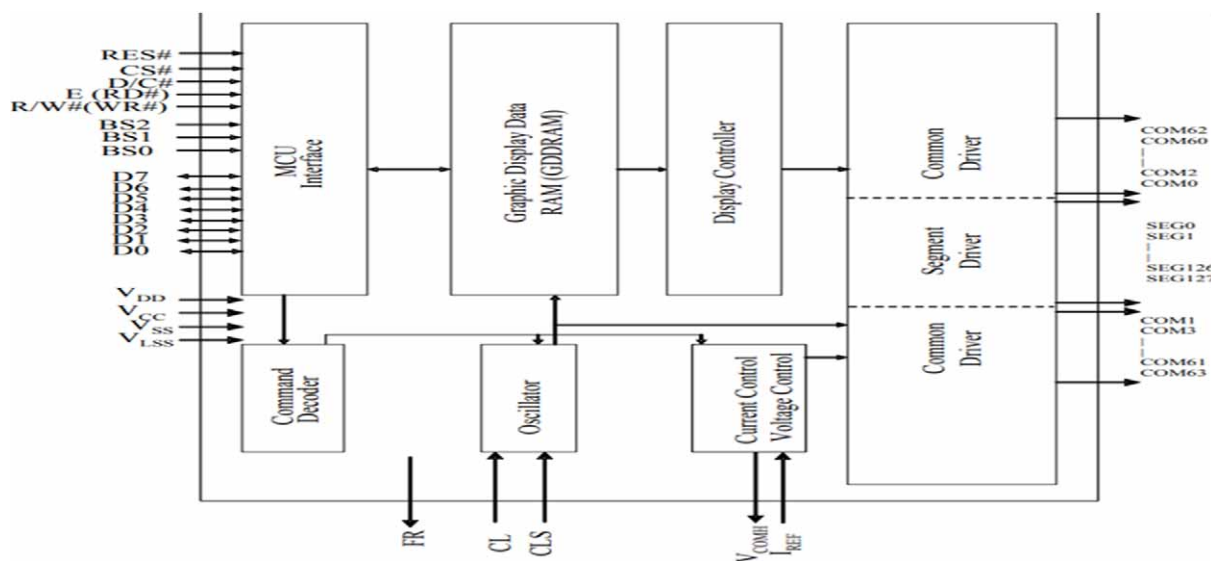


Figure 3.33 Block diagram of SSD1306 OLED Display

3.9.1 Absolute maximum ratings.

Table 3.22 Absolute maximum ratings

Parameter	Symbol	Min	Max	Unit
Supply voltage for logic	VDD	0	4	V
Supply voltage for LCD	VCC	0	15	V
Input voltage	VI	0	VDDIOS	V
Operating temperature	TOP	-40	+80	°C
Storage temperature	TSTG	-40	+80	°C

3.9.2 Electrical characteristics.

Table 3.23 Electrical characteristics

Parameter	Symbol	Condition	Min	Type	Max	Unit
Supply voltage for logic	VDD	---	2.8	3	3.3	V
Supply voltage for display	VCC	-	10	12	15	V
Input high volt	VIH	-	0.8VDD	-	VDDI/O	V
Input low volt	VIL	-	0	-	0.2VDD	V
Output high volt	VOH	-	0.9	-	VDDI/O	V
Output low volt	VOL	---	0	-	0.1	V
Operating current for VCC	ICC	VCC=12V	9	10	12	mA

3.9.3 OLED pin connections.

Table 3.24 OLED Pin connections

Pin No.	Symbol	Description																								
1	NC (GND)	Reserved pins. The reserved pins can reduce the influences from stresses on the function pins. These pins must be connected to external ground																								
2	C2N	Positive terminal of the flying inverting capacitor negative terminal of the flying boost capacitor. The charge pump capacitors required between the terminals																								
3	C2P	Input voltage for LCD																								
4	C1P	H: data signal, L: instruction signal																								
5	C1N	H: read mode, L: write mode																								
6	Vbat	It supplies power for DC/DC converter circuit. It also supplies power for the internal buffer of DC/DC voltage converter.																								
7	NC	NC																								
8	VSS	Ground of logic circuit																								
9	VDD	Power supply of logic circuit																								
10	BS0	These pins are MCU interface selection input.																								
		<table><tr><td></td><td>I²C</td><td>3 wire SPI</td><td>4 wire SPI</td><td>8bit 68xx parallel</td><td>8-bit 80xx parallel</td></tr><tr><td>BS0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>BS1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>BS2</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>		I ² C	3 wire SPI	4 wire SPI	8bit 68xx parallel	8-bit 80xx parallel	BS0	0	1	0	0	0	BS1	1	0	0	0	0	BS2	0	0	0	1	1
			I ² C	3 wire SPI	4 wire SPI	8bit 68xx parallel	8-bit 80xx parallel																			
		BS0	0	1	0	0	0																			
BS1	1	0	0	0	0																					
BS2	0	0	0	1	1																					
11	BS1	Data bit 4																								
12	BS2	Data bit 5																								
13	CS	It works as chip select input. The chip is allowing to work only when it is placed at low.																								
14	RES	Power reset for controller and driver. It works as reset signal input. When the pin is placed at low, it will begin to run																								
15	D/C	Data/Command control. It works as data/command control pin. If the pin is placed high, the input at D1 and D0 is as display data. If the pin is placed low, the input at D1 and D0 will be transferred to the command register.																								
16	R/W	This is MCU interface input. When interfacing to a 68xx series microprocessor, this pin will be used as read/write selection input. Pull this pin too high for read mode and pull it too low for write mode.																								
17	E/RD	It works as MCU interface input. If it is directed to a 68xx microprocessor, this pin will be used as the enable signal. Read/write operation will be executed when placed at high and pin number 13 is low. When connecting to an 80xx microprocessor, this pin receives the read signal.																								
18 to 25	D0 TO D7	These pins are 8-bit two-way directional data bus to be connected to the microprocessors data bus. If we select serial mode, D1 will be serial data input and D0 will be the serial clock input. If we select I ² C mode, then D2 and D1 will be toggle together and serve as SDA out and SDA in.																								
26	Iref	It works as segment current reference pin. A resistor should be connected between this pin and voltage source and current should be lower than 12.5μA.																								
27	Vcomh	Voltage output should be high for communication signal. This pin work as input pin for the output voltage.																								

28	VCC	Power supply for OEL signal. This is the most positive voltage supply pin of the chip.
29	VLSS	Ground of analog circuit. This is an analog ground pin. It should be connected to VSS externally
30	NC (GND)	Reserved pin. The supporting pins can reduce the influxes from stresses on the function pins. These pins must be connected to external ground.

3.10 Power Supply LM1117 module. (800mA Low-Dropout Linear Regulator).

The system power supply design provides energy for the whole system operation, the LM1117 linear regulator main control chip, TFT screen and wireless transmission module to provide a stable 3.3V power supply, for the stability of voltage filtering clutter, in the LM1117 chip at both ports are loaded with an additional set of filter capacitors and decoupling capacitors, LM1117 chip schematic as shown in Figure 3.34

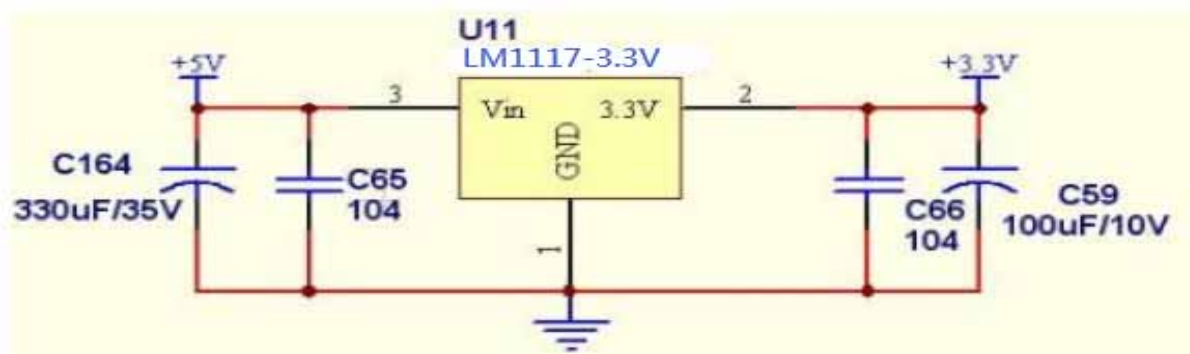


Figure 3 34 LM1117 schematic.

The LM1117 is a popular linear voltage regulator that can provide steady voltage to your Arduino and AT Mega projects. This unit comes in fixed output voltages of 1.8, 2.5, 3.3, and 5 volts or can be bought as an adjustable unit, able to output 1.25 to 13.8 volts. This unit works with an input voltage of up to 20 volts, can handle up to 800ma of current and some versions work down to -40 degrees Celsius, making it ideal for outdoor use or high-altitude balloon flights. This unit also comes with current limiting and thermal protection. The LM1117 is made by Texas Instruments.

3.10.1 General Description of LM1117 module.

The LM1117 is a series of low dropout voltage regulators with a dropout of 1.2V at 800mA of load current. It has the same pin-out as National Semiconductor's industry standard LM317.

The LM1117 is available in an adjustable version, which can set the output voltage from 1.25V to 13.8V with only two external resistors. In addition, it is also available in three fixed voltages, 2.85V, 3.3V, and 5V.

The LM1117 offers current limiting and thermal shutdown. Its circuit includes a Zener trimmed band gap reference to assure output voltage accuracy to within $\pm 1\%$.

The LM1117 series is available in SOT-223 and TO-220 packages. Minimum of 10 μ F tantalum capacitor is required at the output to improve the transient response and stability.

Table 3.25 General Description of LM1117 module.

Output Voltage	Full Device Name	Packaging Details
5.0V	LM1117T-5.0 LM1117MP-5.0 LM1117MPX-5.0	TO-220 SOT-223, 250 units/reel SOT-223, 2000 units/reel
3.3V	LM1117T-3.3 LM1117MP-3.3 LM1117MPX-3.3	TO-220 SOT-223, 250 units/reel SOT-223, 2000 units/reel
2.85V	LM1117T-2.85 LM1117MP-2.85 LM1117MPX-2.85	TO-220 SOT-223, 250 units/reel SOT-223, 2000 units/reel
Adjustable	LM1117T-ADJ LM1117MP-ADJ LM1117MPX-ADJ	TO-220 SOT-223, 250 units/reel SOT-223, 2000 units/reel

3.10.2 Features.

- Available in 2.85V, 3.3V, 5V, and Adjustable Versions
- Space Saving SOT-223 Package
- Current Limiting and Thermal Protection
- Output Current 800mA
- Temperature Range 0°C to 125°C
- Line Regulation 0.2% (Max)
- Load Regulation 0.4% (Max)

3.10.3 Applications.

- 2.85V Model for SCSI-2 Active Termination
- Post Regulator for Switching DC/DC Converter
- High Efficiency Linear Regulators
- Battery Charger
- Battery Powered Instrumentation

Chapter 4 System flow and main function design

4.1 System Flow

4.1.1 Overview of system processes

Turn on the power switch, each sensor and function module in the device is gradually energized, and the main control chip is reset to start running the program after it is powered. In the internal environment, each program begins to initialize, including clock, interrupt, timerization, ad-digital-to-analog converter, spi interface, universal IO port, etc. The various sensor function modules outside the main control have also begun to initialize, such as smoke sensor modules, wireless Bluetooth modules, etc. After the initialization of each functional module is completed and the user system is successfully connected, the cycle detection ends and the device begins to work normally, and some of the flow charts are shown in Figure 4-1.

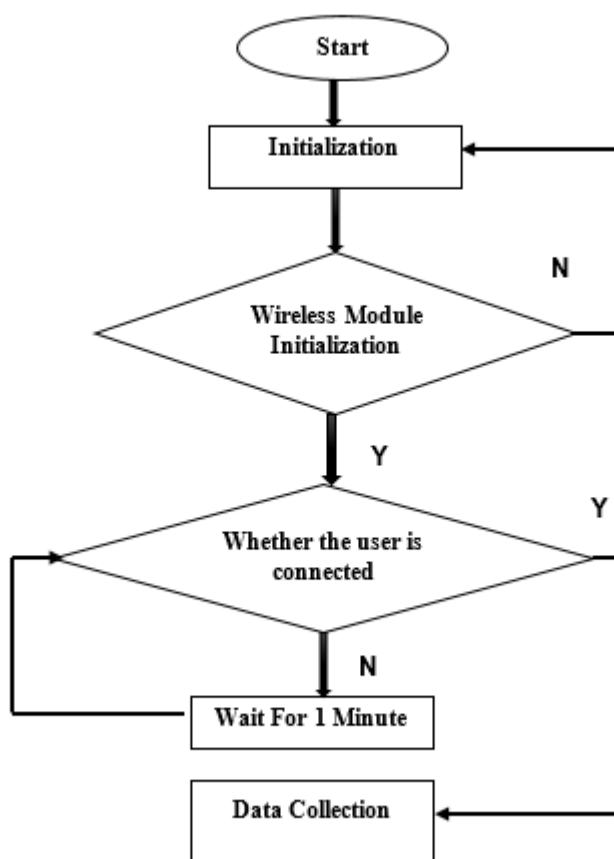


Figure 4-1 Partial system flowchart

The analysis flow chart can clearly see that after the internal resources of each program are initialized, it will be judged whether the wireless module is running normally, and after the normal startup, it will further determine whether the user is successfully connected, and the judgment of this step will be judged by a one-minute waiting cycle until the user successfully accesses. At this time, the cloud platform is officially connected with the next computer, and the functional modules of each sensor begin to collect data in real time. It includes real-time

monitoring of indoor temperature and humidity, indoor gas detection data of smoke sensors, and information of alarm modules. If the alarm module is touched, the master controller will immediately send instructions to the corresponding security processor device, and warn the user, and the user will be alerted through the wireless module, and the user can also view the alarm information record that has occurred in the cloud platform. In the process of program operation, the detection and alarm of various dangerous situations have the highest priority, and then the data transmission of the collected information.

4.1.2 Data Collection Process

The data acquisition process is shown in Figure 4-2, the main controller in the power-up initialization of the pins of each acquisition module, began to perform data collection sequentially, first determine whether the temperature and humidity module is normally initialized, after normal initialization began to collect temperature and humidity, followed by the determination of whether the dangerous gas sensor is triggered, if the trigger jumps to the upload program for warning prompts, if normal then continue to collect data downwards.

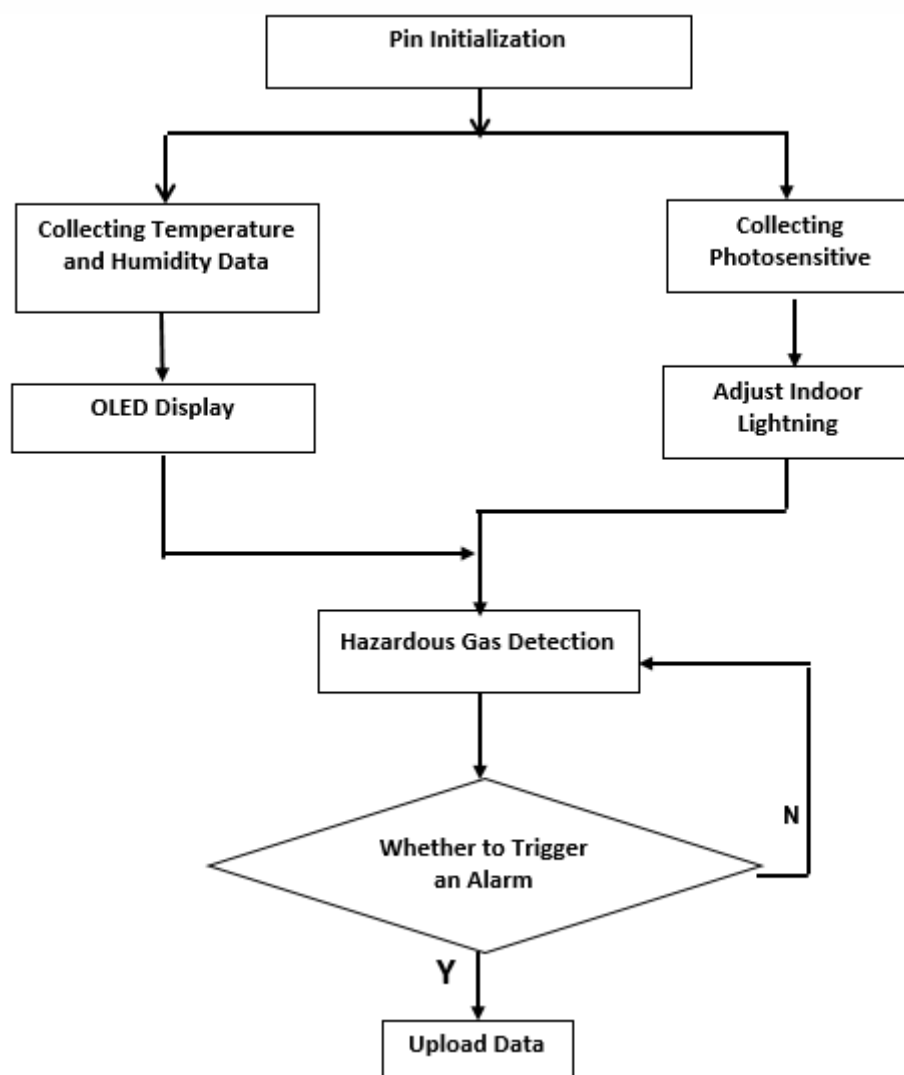


Figure 4 2 Data acquisition flowchart

4.1.3 Cloud Platform Transfer Process

Cloud platform transmission, first write the fixed Wi-Fi network account (ssid) and password (password), establish a penetration network, direct access to the IP address of the cloud server, and use the MQTT protocol to package the transmission data. In this process, the master determines whether to upload the data, if it is determined to upload, upload the collected data, and then wait for the host computer to send the control signal, the transmission process is shown in Figure 4-3.

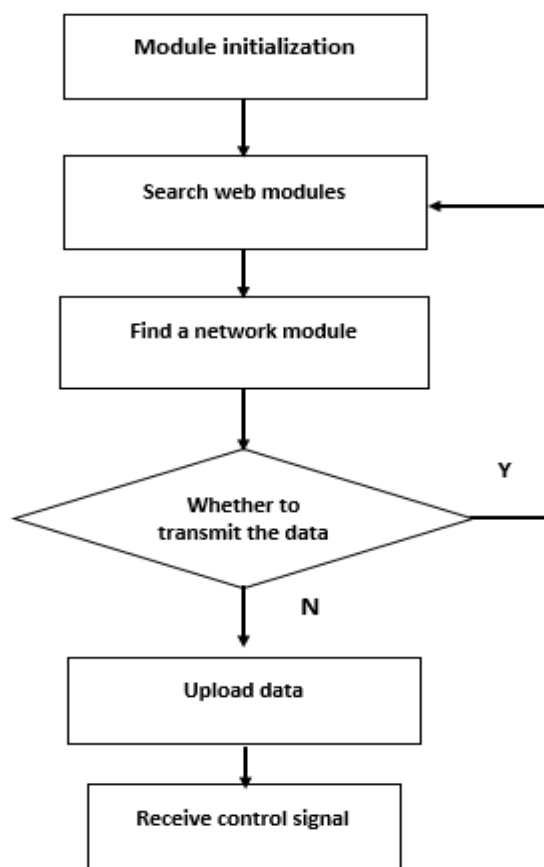


Figure 4 3 Cloud platform flowchart

4.2 Wireless Communication

In the design, the data information collected by each function module will be identified by the master control and sent and uploaded by the ESP8266 wireless Bluetooth module.

Since this wireless module uses serial port communication mode, it is necessary to initialize the serial port and configure the baud rate when debugging the function. It is then functionally set, sending it a "+" to exit the module's default pass-through mode, followed by network configuration and server configuration of the wireless module. After confirming the network name, the network configuration requires the IP address and port number to access the password and the number of users as an administrator.

4.3 Temperature and humidity collection

DHT11 sensor in the acquisition of data information in the environment when the internal resources of the communication mode when single bus communication, this communication mode needs to be in accordance with a certain timing logic sequence to complete the instruction answer. After the module starts to complete the self-test after the response to the initialization of the wait, you can start to collect the temperature and humidity data in the environment, each collection will get a set of 40bit data, the reorganization data is divided into 5 groups to store, each group has 8 bits. Among them, the first and second groups of data are the collected humidity information, the third and fourth groups of data are the temperature information collected by the sensor, and the fifth group is used to verify whether the data is accurate.

4.4 Cloud Platforms

With the help of tools provided by the One NET platform, we can support the construction of individual user platforms to achieve smooth operation of personal or enterprise cloud management devices. The platform provides an open API interface for third-party applications to achieve personalized system construction and can also customize the APP through the platform to further meet the individual needs of different users [17-21].

By connecting the ESP8266 chip of the main control chip STM32 to the external wireless module, the UIP protocol is built on the latter, and then transplanted into the master control chip, the data collected by each sensor during the work process can be uploaded to the cloud platform built. One NET supports HTTP protocol transmission, the main control chip as a mobile client, using protocol request mode, and the cloud platform bidirectional data transmission of various sensor information. As HTTP based on the TCP/IP protocol, the main request types when transmitting data are GET and POST, and when uploading data with the cloud platform, three attributes need to be determined: API key, host, and byte size. As shown in Figure 4-4 POST packets, the byte size of the string data is 63bit, the resource exists in the device's directory, and the device ID is 509226975.

```
POST /devices/509226975/datapoints HTTP/1.1
api-key:aH2orrDNlT1gKXrdurQBoPbm5SI=
Host:api.heclouds.com
Content-Length:63

{"datastreams":[{"id":"sys_time","datapoints":[{"value":50}]}]}
```

Figure 4 4 POST packets

The data is uploaded to the cloud platform through the main control chip, the flag bit is sent after the timer is turned on, and the waiting state machine is set, and the collected information is packaged into the data in the above format, uploaded to the cloud platform in post format, and then the user can observe the relevant state in the interface.

Chapter 5 System debugging

5.1 Hardware debugging

Considering the environment of development and production, this design chose to use DXP software to draw electronic PCB board printing production, in the drawing process, first of all, the main control chip module is drawn, which is also the reason, ignoring the important influence and role of power supply design in the circuit, resulting in unreasonable power design in the late design of the design, so as to draw the PCB board and secondary printing for the second time, delaying the debugging arrangement of one week. After the design of the power module is completed, the multimeter is used to detect the circuit on and off, whether there is a short circuit or a broken circuit, and then the power-on test is carried out to detect whether there is a voltage at each point and whether the voltage value meets the design requirements. The next step is to weld the main control chip and the display screen, in the welding process, pay special attention to the pin position of the chip, while avoiding direct contact between the soldering iron and the two modules, so as not to burn the chip at high temperature. The remaining modules can be tested using in-line access circuitry for subsequent adjustments and data testing.

After the completion of the welding work of each module, it is necessary to use a multimeter to short-circuit test and open circuit test on each solder joint, so as to avoid the virtual welding in the circuit and other situations affecting the subsequent functional test, and the pin voltage of each chip should be measured and recorded to ensure that the connection between VCC and GND is correct, so as not to burn the chip or other devices after re-energization. After checking the safety of the basic circuit, you can officially power up the circuit, such as no abnormal phenomena or odors after power-on, you can start debugging the software.

During commissioning, the physical connection is the A11 pin connecting DHT11 to the core controller, the ESP8266 to the B2 and B3 pins of the core controller, and the A9 and A10 pins of the core controller connected to the CH340 serial port debugging module. The physical connection diagram is shown in Figure 5-1 below:

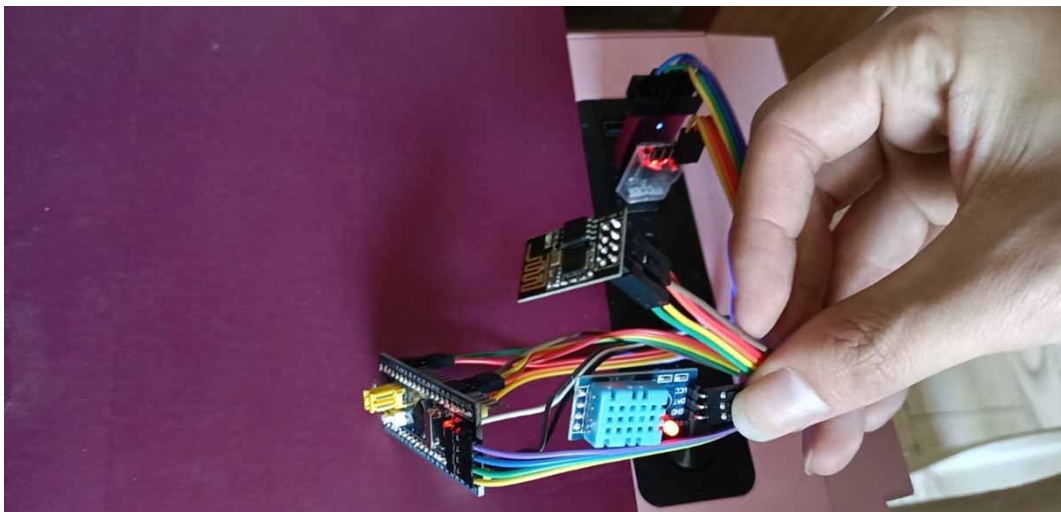


Figure 5 1 Physical connection diagram

5.2 Software debugging

Each functional module is connected to the main control in turn, the module is tested one by one, the data adjustment and transmission confirmation are carried out, and after all the access, the multiple functions are jointly tuned, the program and timing logic are adjusted, and different transmission cycles are arranged according to different data sizes to ensure that multiple functions and multiple data transmission can be run smoothly, and there will be no data confusion. After the overall function can be realized, it is then connected to the cloud platform to test the wireless transmission function, so as to experiment with the master control and the cloud platform and can achieve stable data transmission and control instruction execution on each function.

The cloud platform function is implemented by Alibaba cloud platform, first of all, you need to register a personal user account on the Alibaba cloud platform, and you can carry out product development under the personal account; after entering the console, select the development of the Internet of Things platform, create a new project, name it the smart home cloud platform, configure the node type of the development to be the device, set the networking mode to 2G/4G after determining the project product as shown in Figure 5-2 below:



Figure 5 2 Cloud platform generates an operation diagram

Enter the product, you need to add product functions, take temperature and humidity as an example, if there is no option, you can click other types, search, add product functions, and add as shown in Figure 5-3 below:



Figure 5-3 Cloud platform product function addition diagram

After adding the functions of the cloud platform product, the panel configuration needs to be carried out, and the panel control that adds temperature and humidity directly drags and drops the layout, as shown in Figure 5-4:

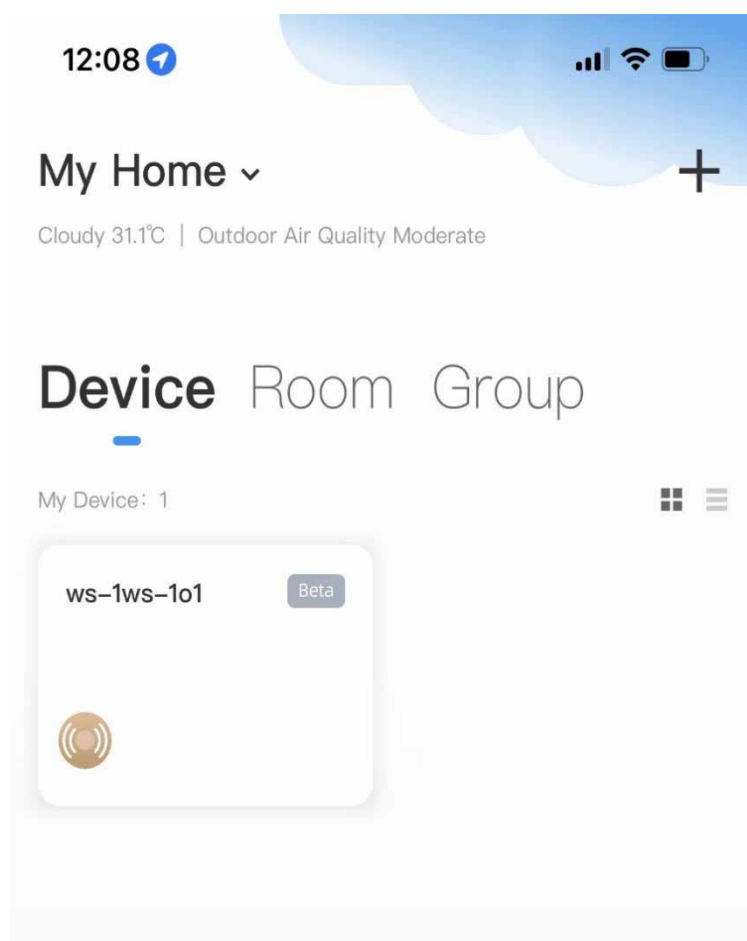


Figure 5-4 Panel Control Layout Diagram

After the panel completes the layout, the next step is carried out: generate the channel of device connection; select the module of the connected device, the ESP8266 used this time is the same parameter as the 202 modules of Hezhou, which can be universal, so the 202 modules of Hezhou is directly used. After determining the module, the following figure 5-5 is generated:

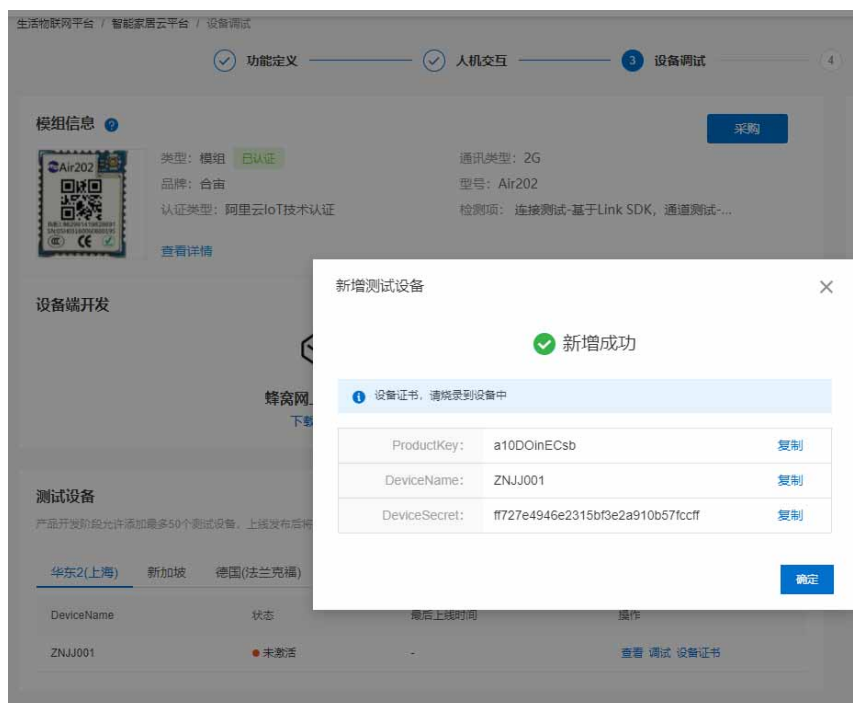


Figure 5 5 Added equipment test diagram

As can be seen from Figure 5-5, the newly established device KEY, driver name, and password string are:

Product Key:

a10DOInECsb

Device Name:

ZNJJ001

Device Secret:

ff727e4946e2315bf3e2a910b57fccff

Configure the above three strings of characters into the development program of STM32 to achieve interconnection. The STM32 program also requires the account number and password configured for the home wireless network.

Serial port debugging to observe whether the data is successfully uploaded, as shown in Figure 5-6 below:

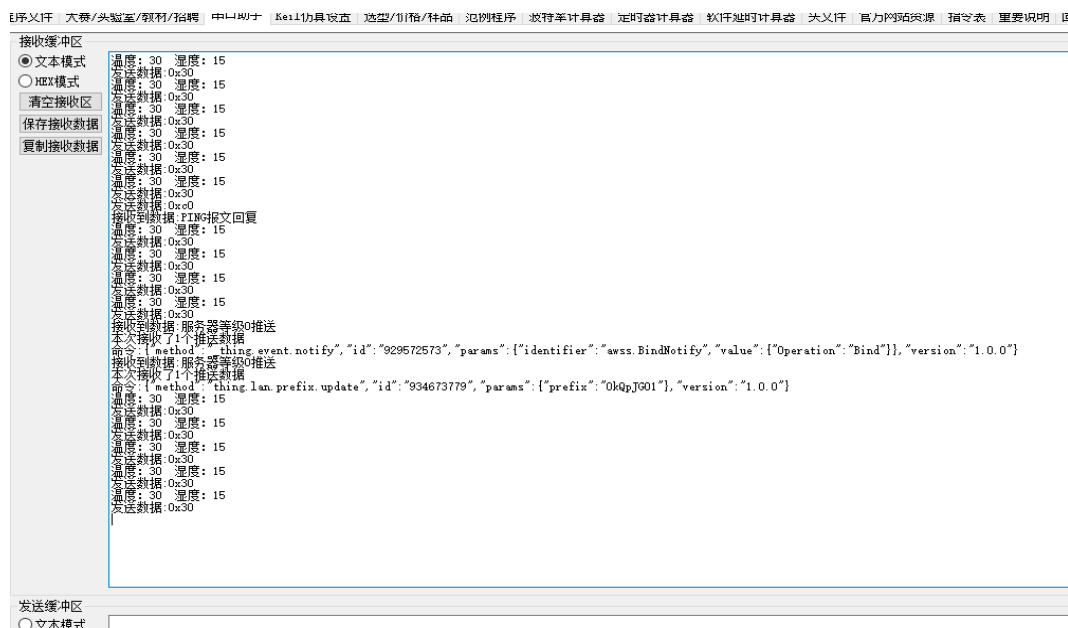


Figure 5 6 Serial port debugging data diagram

Compare whether the serial port tune tries to match the data obtained by the mobile device from the cloud, as shown in Figure 5-7 below:



Figure 5 7 Mobile device data view diagram

After comparison, the temperature and humidity data are completely collected on the STM32 and transmitted to the cloud through esp8266, and the data is successfully read by the mobile terminal.

Conclusion

After several months of hard work, under the guidance of the teacher, the graduation project was successfully completed. The difficulties in the design process exist in the selection of the device and the later debugging process, in the early stage of the work, the selected circuit design scheme leads to a larger impact between the modules with larger power, and the communication mode and transmission characteristics of the cloud platform are unfamiliar. Through the appropriate adjustment of the temperature and humidity sensor and the light sensor, the difference obtained by the standard measuring instrument is measured many times, and the algorithm is used to compensate, and the sensitivity and accuracy of this function of the work are successfully stabilized within expectations.

For the use of cloud platforms, I only used the modular experience I had been exposed to before to apply them, but I could not achieve the expected functions well. Through reading the developer manual, I have a certain grasp of the different functions of the cloud platform, conduct some targeted tests according to the functions to be implemented by the design, fully analyze the data and phenomena, and finally choose to use the One Net cloud platform to transmit in the form of HTTP. HTTP is based on the TCP / IP protocol data is simple and reliable, is an important reason to choose it, in the data packaging process will not cause too much load on the system, so that the response time of each data transmission is very fast, and the master chip for data feedback processing speed is also in line with the design of the expected goals.

After solving the debugging problem, the information acquisition of the peripheral functional circuit of the core controller is preliminarily completed, and the data transmission of each module and the master control is analyzed and processed in the follow-up debugging, and the data acquisition of the cloud network is observed in the user interface, and finally the personal user can realize real-time monitoring and effective control remotely.

References

1. depts.washington.edu/dmgftp/publications/html/smarthouse98-mdg.html.
2. Nicoleta Arghira, Lamis Hawarah, Stéphane Ploix, and Mireille Jacomino, "Prediction of Appliances Energy Use in Smart Homes," *Energy*, vol. 48, Iss. 1, pp. 128- 134, Dec. 2012.
3. Gayathri N., Vineeth V.V., and N. Radhika, "A Novel Approach in Demand Side Management for Smart Home," *Procedia Technology*, Elsevier Ltd., vol. 21, pp. 526-532, 2015.
4. Prashant Kumar, Claudio Martani, Lidia Morawska, Leslie Norford, Ruchi Choudhary, Margaret Bell, and Matt Leach, "Indoor Air Quality and Energy Management Through Real-Time Sensing in Commercial Buildings," *Energy and Buildings*, vol. 111, pp. 145-153, Jan. 2016.
5. Mohammad Ali Fotouhi Ghazvini, João Soares, Omid Abrishambaf, Rui Castro, and Zita Vale, "Demand Response Implementation in Smart Households," *Energy and Buildings*, vol. 143, pp.129-148, May 2017.
6. Yang, Shuang-Hua, "ZigBee Smart Home Automation Systems-Wireless Sensor Networks: Principles, Design and Applications," Springer-Verlag London 2014.
7. Amirah Aisha Badrul Hishama, Mohamad Hafis Izran Ishaka, Chan Kok Teika, Zaharuddin Mohameda, and Nurul Hawani Idrisb, "Bluetooth-Based Home Automation System Using an Android Phone," *Jurnal Teknologi (Sciences & Engineering)*, Vol.70, No.3, pp.57–61, 2014.
8. Bader M. O. Al-thobaiti, Iman I. M. Abosolaiman, Mahdi H. M. Alzahrani, Sami H. A. Almalki, and Mohamed S. Soliman "Design and Implementation of a Reliable Wireless Real-Time Home Automation System Based on Arduino Uno Single-Board Microcontroller" *International Journal of Control, Automation and Systems* vol. 3, no.3, pp. 11-15, Jul. 2014.
9. Iman I. M. Abu Sulayman, Sami H. A. Almalki, and Mohamed S. Soliman "Designing a Reliable Dual Modes Real-Time Home Automation System Based on Very High Speed Description Language" *International Journal of Control, Automation and Systems* vol. 5, no.3, pp. 19-23, Jul. 2016.
10. *Home Automation & Wiring* (1 ed.). New York: McGraw-Hill/TAB Electronics. 1999- 03-31. ISBN 978-0-07-024674-4.
11. Anitha, A. (2017, November). Home security system using internet of things. In *IOP conference series: materials science and engineering* (Vol. 263, No. 4, p. 042026). IOP Publishing.
12. Anitha, A. "Home security system using internet of things." *IOP conference series: materials science and engineering*. Vol. 263. No. 4. IOP Publishing, 2017.
13. Informatika, T. et al. (nodate) 'Sistem Monitoring Suhu Dan Kelembaban Ruang Produksi Berbasis Wireless Sensor Network Pada Pt . Xxx Manufacturing Services Indonesia', pp. 136–143.

14. Pooja Ajmera, 2017, A Review Paper on Infrared sensor, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) VIMPACT – 2017 (Volume 5 – Issue 23),.
15. K. Bankapur, H. Mathur, H. Singh, R. Harikrishnan and A. Gupta, "A Flame Sensor-Based Firefighting Assistance Robot with Simulation Based Multi-Robot Implementation," 2022 *International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI)*, 2022, pp. 1-8, doi: 10.1109/ACCAI53970.2022.9752470.
16. Ramya, V., & Palaniappan, B. (2012). Embedded system for Hazardous Gas detection and Alerting. *International Journal of Distributed and Parallel Systems (IJDPS)*, 3(3), 287-300.
17. David, N., Chima, A., Ugochukwu, A., & Obinna, E. (2015). Design of a home automation system using arduino. *International Journal of Scientific & Engineering Research*, 6(6), 795-801.
18. Answers.Com/Sonar/En/Com.
19. St.Com/Resource/En/Datasheet/Stm32f103c8.Pdf.

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Academic Publication Highlights:

<https://www.airitilibrary.com/Publication/alDetailedMesh?docid=P20190813001-202201-202112290001-202112290001-536-541>

The Design of STM 32 Micro-Controller Intelligent Home System based on Cloud Platform

Adnan Abdullah ; Shuang Wang ; Yihang Cai

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英文 DOI: 10.6919/IJCE.202201_8(1).0074 ?

ARM Cortex-M3 STM32 Micro-Controller ; Temperature and Humidity Sensor ; Smoke Sensor ; Wireless Transmission ; Flame Sensor ; Buzzer

Appendix:

The device KEY, driver name, and password string are:

Product Key:

a10DOinECsb

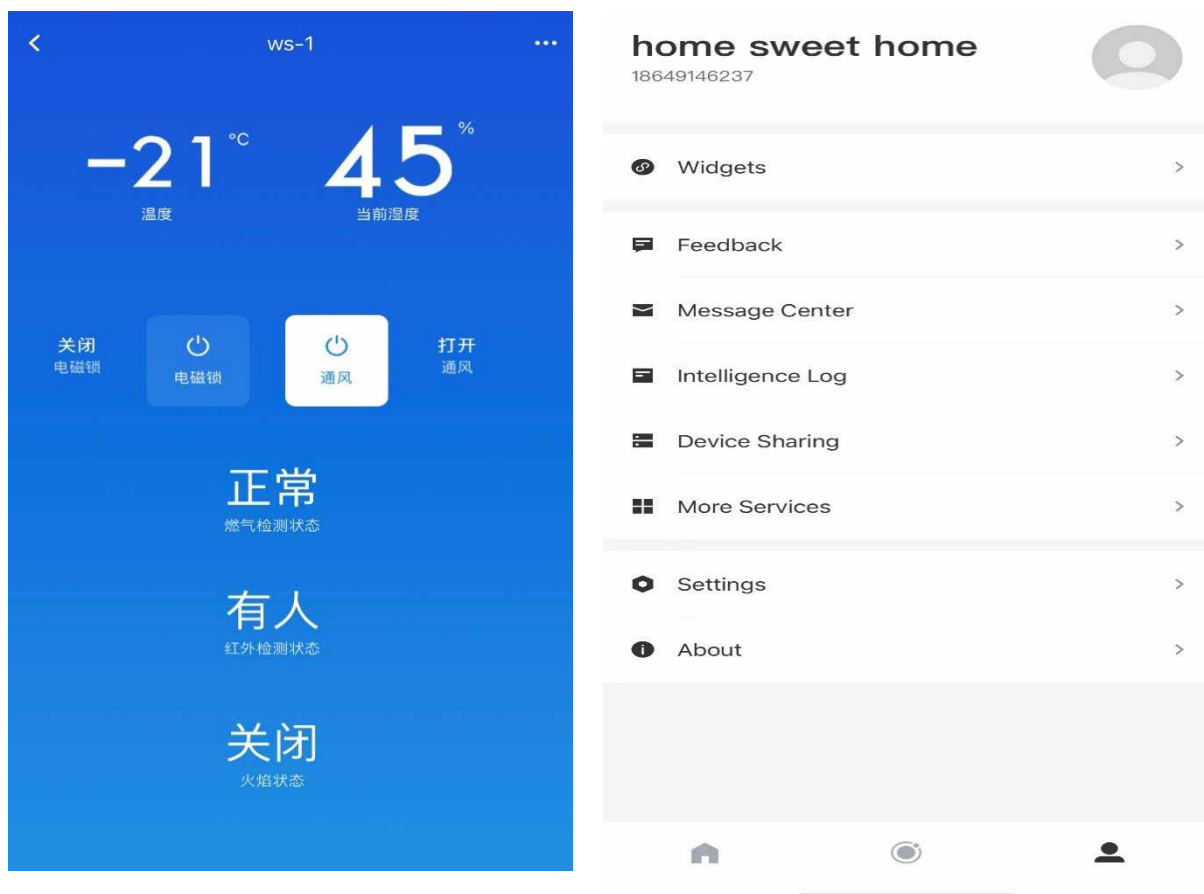
Device Name:

ZNJJ001

Device Secret:

ff727e4946e2315bf3e2a910b57fccff

On the device



After adding the functions of the cloud platform product, the panel configuration needs to be conducted, and the panel control that adds temperature and humidity directly drags and drops the layout, as shown in Figure

